

Scaling Laws for Neural Language Models

神经语言模型的 *scaling laws*

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Abstract / 摘要

EN We study empirical scaling laws for language model performance on the cross-entropy loss.

中 我们研究语言模型在交叉熵损失上的性能所遵循的经验 *scaling laws*。

EN The loss scales as a power-law with model size, dataset size, and the amount of compute used for training, with some trends spanning more than seven orders of magnitude.

中 损失随模型规模、数据集大小和训练所用计算量呈幂律缩放，其中一些趋势横跨七个以上的数量级。

EN Other architectural details such as network width or depth have minimal effects within a wide range.

中 在很宽的范围内，网络宽度或深度等其他架构细节的影响很小。

EN Simple equations govern the dependence of overfitting on model/dataset size and the dependence of training speed on model size.

中 简单的方程刻画了过拟合对模型/数据集大小的依赖关系，以及训练速度对模型规模的依赖关系。

EN These relationships allow us to determine the optimal allocation of a fixed compute budget.

中 这些关系使我们能够确定固定计算预算的最优分配方式。

EN Larger models are significantly more sample-efficient, such that optimally compute-efficient training involves training very large models on a relatively modest amount of data and stopping significantly before convergence.

中 更大模型的样本效率显著更高，因此，计算效率最优的训练方式是在相对适量的数据上训练非常大的模型，并在远未收敛时就停止训练。

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1 Introduction

引言

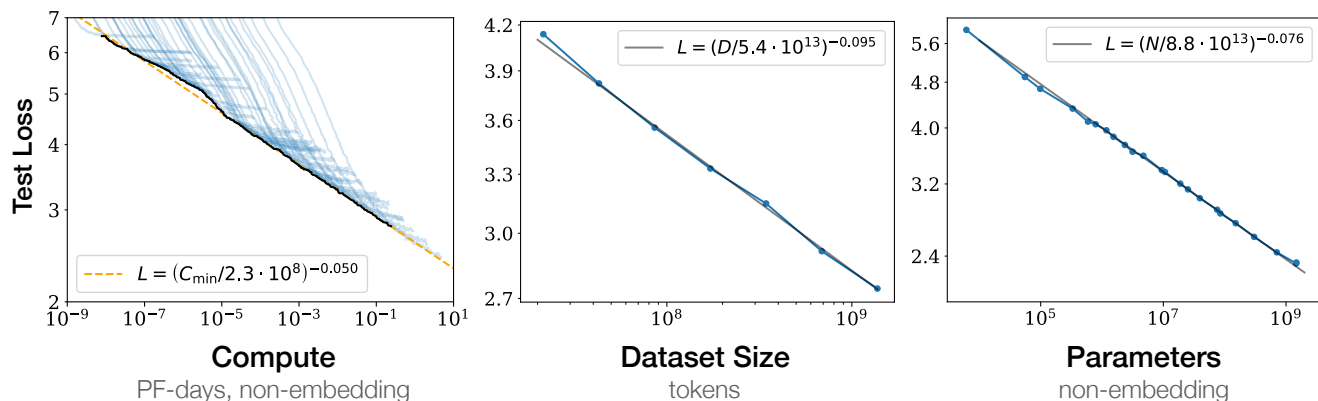


Figure / 图 1 *EN* Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute¹ used for training.

中 随着模型规模、数据集大小和训练所用计算量¹的增加，语言建模性能会平滑提升。

EN For optimal performance all three factors must be scaled up in tandem.

中 为获得最优性能，必须同步扩大这三个因素。

EN Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

中 当性能不受另外两个因素制约时，经验性能与每个单独因素之间都呈幂律关系。

EN Language provides a natural domain for the study of artificial intelligence, as the vast majority of reasoning tasks can be efficiently expressed and evaluated in language, and the world’s text provides a wealth of data for unsupervised learning via generative modeling.

中 语言为人工智能研究提供了一个自然的领域，因为绝大多数推理任务都能以语言高效地表达和评估，而世界上的文本也为通过生成式建模进行无监督学习提供了丰富的数据。

EN Deep learning has recently seen rapid progress in language modeling, with state of the art models [RNSS18, DCLT18, YDY⁺19, LOG⁺19, RSR⁺19] approaching human-level performance on many specific tasks [WPN⁺19], including the composition of coherent multi-paragraph prompted text samples [RWC⁺19].

中 最近，深度学习在语言建模方面取得了迅速进展，最先进的模型 [RNSS18, DCLT18, YDY⁺19, LOG⁺19, RSR⁺19] 在许多具体任务上正接近人类水平的性能 [WPN⁺19]，其中包括生成连贯的多段落提示文本样本 [RWC⁺19]。

EN One might expect language modeling performance to depend on model architecture, the size of neural models, the computing power used to train them, and the data available for this training process.

中 人们或许会预期，语言建模性能取决于模型架构、神经模型的规模、用于训练模型的计算能力，以及可供这一训练过程使用的数据。

EN In this work we will empirically investigate the dependence of language modeling loss on all of these factors, focusing on the Transformer architecture [VSP⁺17, LSP⁺18].

中 在本研究中，我们将从经验上探究语言建模损失对所有这些因素的依赖关系，并重点关注 Transformer 架构 [VSP⁺17, LSP⁺18]。

EN The high ceiling and low floor for performance on language tasks allows us to study trends over more than seven orders of magnitude in scale.

中 语言任务性能的上限很高、下限很低，这使我们能够研究尺度横跨七个以上数量级的趋势。

EN Throughout we will observe precise power-law scalings for performance as a function of training time, context length, dataset size, model size, and compute budget.

中 在全文中，我们将观察到性能随训练时间、上下文长度、数据集大小、模型规模和计算预算变化的精确幂律缩放关系。

1.1 Summary 总结

¹*EN* Here we display predicted compute when using a sufficiently small batch size.

中 这里展示的是使用足够小的 batch size 时预测的计算量。

EN See Figure 13 for comparison to the purely empirical data.

中 与纯经验数据的比较见图 13。

EN Our key findings for Transformer language models are as follows:

中 我们对 Transformer 语言模型的主要发现如下：

EN Performance depends strongly on scale, weakly on model shape: Model performance depends most strongly on scale, which consists of three factors: the number of model parameters N (excluding embeddings), the size of the dataset D , and the amount of compute C used for training.

中 性能强烈依赖尺度，而很少依赖模型形状：模型性能最强烈地依赖于尺度，尺度由三个因素构成：模型参数数量 N (不包括 embeddings)、数据集大小 D ，以及训练所用的计算量 C 。

EN Within reasonable limits, performance depends very weakly on other architectural hyperparameters such as depth vs. width.

中 在合理范围内，性能对深度与宽度等其他架构超参数的依赖非常弱。

EN (Section 3)

中 (第 3 节)

EN Smooth power laws: Performance has a power-law relationship with each of the three scale factors N, D, C when not bottlenecked by the other two, with trends spanning more than six orders of magnitude (see Figure 1).

中 平滑的幂律：当性能不受另外两个因素制约时，它与三个尺度因素 N, D, C 中的每一个都呈幂律关系，相关趋势横跨六个以上的数量级（见图 1）。

EN We observe no signs of deviation from these trends on the upper end, though performance must flatten out eventually before reaching zero loss.

中 在高端，我们没有观察到偏离这些趋势的迹象，不过性能最终必然会在达到零损失之前趋于平坦。

EN (Section 3)

中 (第 3 节)

EN Universality of overfitting: Performance improves predictably as long as we scale up N and D in tandem, but enters a regime of diminishing returns if either N or D is held fixed while the other increases.

中 过拟合的普适性：只要同步扩大 N 和 D ，性能就会以可预测的方式提升；但如果固定 N 或 D 中的一个而增大另一个，就会进入收益递减的区域。

EN The performance penalty depends predictably on the ratio $N^{0.74}/D$, meaning that every time we increase the model size 8x, we only need to increase the data by roughly 5x to avoid a penalty.

中 性能损失以可预测的方式取决于比率 $N^{0.74}/D$ ，这意味着模型规模每扩大 8 倍，只需将数据增加约 5 倍即可避免性能损失。

EN (Section 4)

中 (第 4 节)

EN Universality of training: Training curves follow predictable power-laws whose parameters are roughly independent of the model size.

中 训练的普适性：训练曲线遵循可预测的幂律，其参数大致不依赖于模型规模。

EN By extrapolating the early part of a training curve, we can roughly predict the loss that would be achieved if we trained for much longer.

中 通过外推训练曲线的早期部分，我们可以粗略预测，如果继续训练很长时间，模型将达到怎样的损失。

EN (Section 5)

中 (第 5 节)

EN Transfer improves with test performance: When we evaluate models on text with a different distribution than they were trained on, the results are strongly correlated to those on the training validation set with a roughly constant offset in the loss – in other words, transfer to a different distribution incurs a constant penalty but otherwise improves roughly in line with performance on the training set.

中 迁移能力随测试性能提升：当我们在与训练文本分布不同的文本上评估模型时，结果与训练分布验证集上的结果高度相关，二者的损失之间存在大致恒定的偏移——换言之，迁移到不同分布会带来恒定的性能损失，但除此之外，其改善程度大体与训练集上的性能同步。

EN (Section 3.2.2)

中 (第 3.2.2 节)

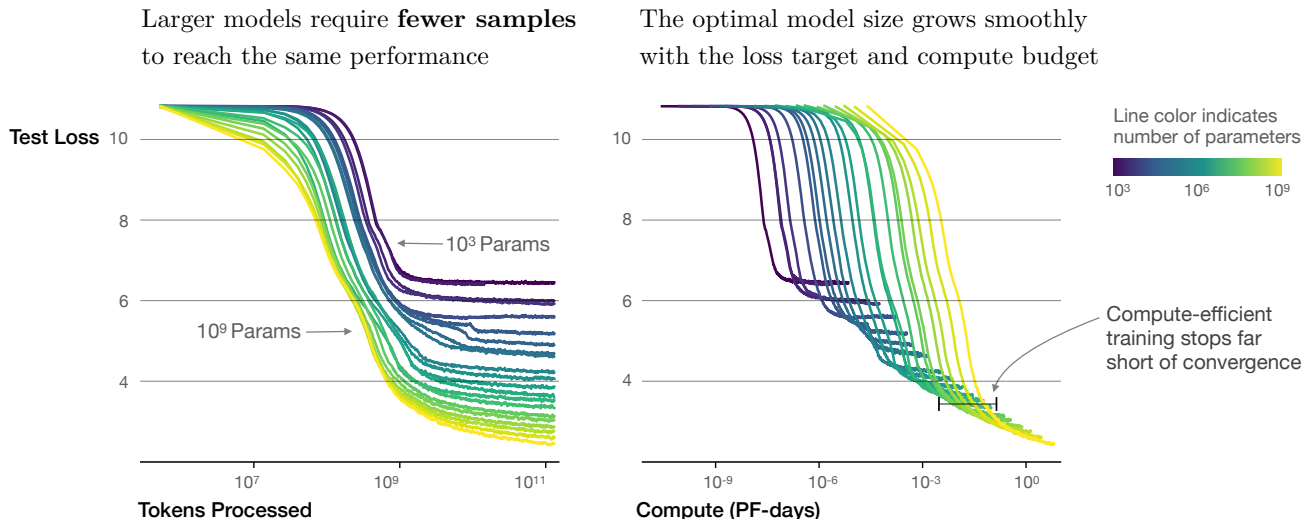


Figure / 图 2 *EN* We show a series of language model training runs, with models ranging in size from 10^3 to 10^9 parameters (excluding embeddings).

中 我们展示了一系列语言模型训练过程，模型规模从 10^3 到 10^9 个参数（不包括 embeddings）不等。

EN **Sample efficiency:** Large models are more sample-efficient than small models, reaching the same level of performance with fewer optimization steps (Figure 2) and using fewer data points (Figure 4).

中 样本效率：大模型的样本效率比小模型更高，使用更少的优化步骤（图 2）和更少的数据点（图 4）即可达到相同的性能水平。

EN **Convergence is inefficient:** When working within a fixed compute budget C but without any other restrictions on the model size N or available data D , we attain optimal performance by training *very large models* and stopping *significantly short of convergence* (see Figure 3).

中 训练至收敛是低效的：当我们在固定计算预算 C 内工作，但模型规模 N 或可用数据 D 不受任何其他限制时，可以通过训练非常大的模型并在远未收敛时停止来获得最优性能（见图 3）。

EN Maximally compute-efficient training would therefore be far more sample efficient than one might expect based on training small models to convergence, with data requirements growing very slowly as $D \sim C^{0.27}$ with training compute.

中 因此，相较于根据小模型训练至收敛所得出的预期，计算效率最高的训练会具有高得多的样本效率；其数据需求随训练计算量仅以 $D \sim C^{0.27}$ 的速度非常缓慢地增长。

EN (Section 6)

中 （第 6 节）

EN **Optimal batch size:** The ideal batch size for training these models is roughly a power of the loss only, and continues to be determinable by measuring the gradient noise scale [MKAT18]; it is roughly 1-2 million tokens at convergence for the largest models we can train.

中 最优 **batch size**：训练这些模型的理想 batch size 大致仅为损失的幂函数，并且仍然可以通过测量 gradient noise scale [MKAT18] 来确定；对于我们能够训练的最大模型，收敛时的理想 batch size 约为 100 万至 200 万个 tokens。

EN (Section 5.1)

中 （第 5.1 节）

EN Taken together, these results show that language modeling performance improves smoothly and predictably as we appropriately scale up model size, data, and compute.

中 综合来看，这些结果表明，只要适当地扩大模型规模、数据和计算量，语言建模性能就会平滑且可预测地提升。

EN We expect that larger language models will perform better and be more sample efficient than current models.

中 我们预计，更大的语言模型将比当前模型表现得更好，并且样本效率更高。

1.2 Summary of Scaling Laws

Scaling Laws 总结

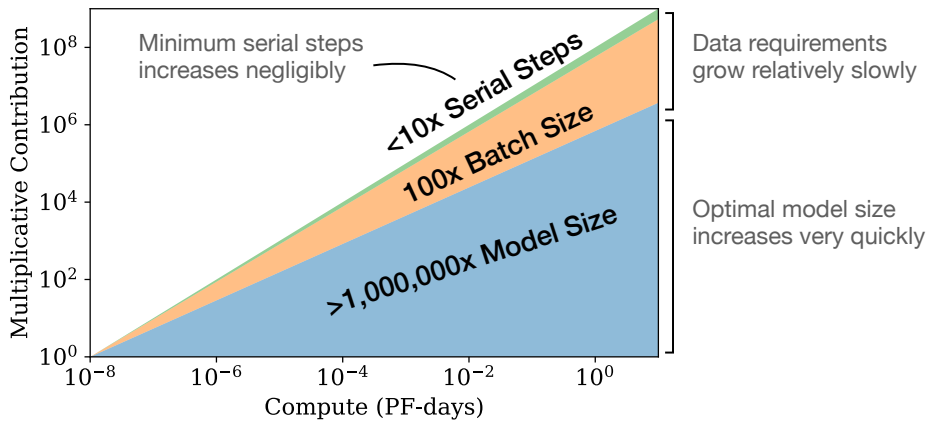


Figure / 图 3 **EN** As more compute becomes available, we can choose how much to allocate towards training larger models, using larger batches, and training for more steps.

中 随着可用计算量增加，我们可以选择分配多少计算量来训练更大的模型、使用更大的批量，以及训练更多的步骤。

EN We illustrate this for a billion-fold increase in compute.

中 这里以计算量增加十亿倍为例进行说明。

EN For optimally compute-efficient training, most of the increase should go towards increased model size.

中 对于计算效率最优的训练，大部分增量都应分配给模型规模的扩大。

EN A relatively small increase in data is needed to avoid reuse.

中 只需相对少量地增加数据即可避免数据复用。

EN Of the increase in data, most can be used to increase parallelism through larger batch sizes, with only a very small increase in serial training time required.

中 在增加的数据中，大部分可以通过更大的 batch size 用于提高并行度，而所需的串行训练时间只需增加非常少。

EN The test loss of a Transformer trained to autoregressively model language can be predicted using a power-law when performance is limited by only either the number of non-embedding parameters N , the dataset size D , or the optimally allocated compute budget $C_{\min}$ (see Figure 1):

中 对于经过训练、以 autoregressive 方式对语言建模的 Transformer，当性能只受 non-embedding parameters 的数量 N 、数据集大小 D 或经过最优分配的计算预算 $C_{\min}$ 三者之一限制时，其测试损失可以用幂律预测（见图 1）：

1. **EN** For models with a limited number of parameters, trained to convergence on sufficiently large datasets:

中 对于参数数量有限、在足够大的数据集上训练至收敛的模型：

$$L(N) = (N_c/N)^{\alpha_N}; \quad \alpha_N \sim 0.076, \quad N_c \sim 8.8 \times 10^{13} \text{ (non-embedding parameters)} \quad (1.1)$$

2. **EN** For large models trained with a limited dataset with early stopping:

中 对于使用有限数据集并通过 early stopping 进行训练的大模型：

$$L(D) = (D_c/D)^{\alpha_D}; \quad \alpha_D \sim 0.095, \quad D_c \sim 5.4 \times 10^{13} \text{ (tokens)} \quad (1.2)$$

3. **EN** When training with a limited amount of compute, a sufficiently large dataset, an optimally-sized model, and a sufficiently small batch size (making optimal² use of compute):

中 当使用有限的计算量、足够大的数据集、规模最优的模型以及足够小的 batch size 进行训练时（从而最优地利用³计算量）：

$$L(C_{\min}) = (C_c^{\min}/C_{\min})^{\alpha_C^{\min}}; \quad \alpha_C^{\min} \sim 0.050, \quad C_c^{\min} \sim 3.1 \times 10^8 \text{ (PF-days)} \quad (1.3)$$

²We also observe an empirical power-law trend with the training compute C (Figure 1) while training at fixed batch size, but it is the trend with $C_{\min}$ that should be used to make predictions. They are related by equation (5.5).

³在固定 batch size 下进行训练时，我们也观察到训练计算量 C （图 1）所遵循的经验幂律趋势，但进行预测时应使用 $C_{\min}$ 所遵循的趋势。二者通过方程 (5.5) 联系起来。

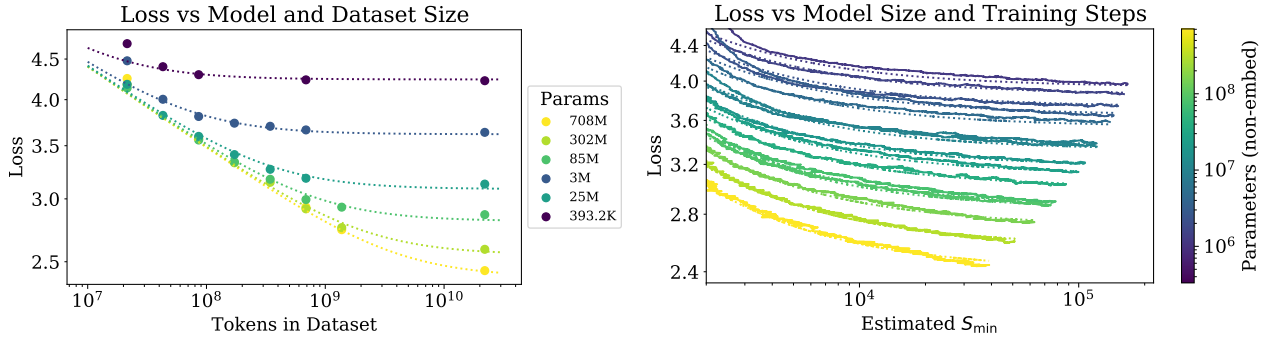


Figure / 图 4 **EN** Left: The early-stopped test loss $L(N, D)$ varies predictably with the dataset size D and model size N according to Equation (1.5).

中 左图：根据方程 (1.5)，early-stopped 测试损失 $L(N, D)$ 随数据集大小 D 和模型规模 N 以可预测的方式变化。

EN Right: After an initial transient period, learning curves for all model sizes N can be fit with Equation (1.6), which is parameterized in terms of $S_{\min}$, the number of steps when training at large batch size (details in Section 5.1).

中 右图：在最初的瞬态阶段之后，所有模型规模 N 的学习曲线都可以用方程 (1.6) 拟合；该方程以 $S_{\min}$ 为参数，后者是在大 batch size 下训练时的步骤数（详见第 5.1 节）。

EN These relations hold across eight orders of magnitude in $C_{\min}$, six orders of magnitude in N , and over two orders of magnitude in D .

中 这些关系在 $C_{\min}$ 横跨八个数量级、 N 横跨六个数量级、 D 横跨两个以上数量级的范围内均成立。

EN They depend very weakly on model shape and other Transformer hyperparameters (depth, width, number of self-attention heads), with specific numerical values associated with the Webtext2 training set [RWC⁺19].

中 它们对模型形状和其他 Transformer 超参数（深度、宽度、自注意力头数量）的依赖非常弱，其中的具体数值与 Webtext2 训练集 [RWC⁺19] 有关。

EN The power laws $\alpha_N, \alpha_D, \alpha_C^{\min}$ specify the degree of performance improvement expected as we scale up N, D , or $C_{\min}$; for example, doubling the number of parameters yields a loss that is smaller by a factor $2^{-\alpha_N} = 0.95$.

中 幂律 $\alpha_N, \alpha_D, \alpha_C^{\min}$ 指定了在扩大 N, D 或 $C_{\min}$ 时预期的性能提升程度；例如，将参数数量加倍，会使损失缩小至原来的 $2^{-\alpha_N} = 0.95$ 倍。

EN The precise numerical values of $N_c, C_c^{\min}$, and D_c depend on the vocabulary size and tokenization and hence do not have a fundamental meaning.

中 $N_c, C_c^{\min}$ 和 D_c 的精确数值取决于词表大小和 tokenization 方式，因此不具有根本意义。

EN The critical batch size, which determines the speed/efficiency tradeoff for data parallelism ([MKAT18]), also roughly obeys a power law in L :

中 决定数据并行中速度/效率权衡的 critical batch size ([MKAT18]) 也大致服从关于 L 的幂律：

$$B_{\text{crit}}(L) = \frac{B_*}{L^{1/\alpha_B}}, \quad B_* \sim 2 \cdot 10^8 \text{ tokens}, \quad \alpha_B \sim 0.21 \quad (1.4)$$

EN Equation (1.1) and (1.2) together suggest that as we increase the model size, we should increase the dataset size sublinearly according to $D \propto N^{\frac{\alpha_N}{\alpha_D}} \sim N^{0.74}$.

中 方程 (1.1) 和 (1.2) 共同表明，随着模型规模增加，我们应按照 $D \propto N^{\frac{\alpha_N}{\alpha_D}} \sim N^{0.74}$ 以次线性方式增大数据集。

EN In fact, we find that there is a single equation combining (1.1) and (1.2) that governs the simultaneous dependence on N and D and governs the degree of overfitting:

中 事实上，我们发现，存在一个将 (1.1) 和 (1.2) 结合起来的单一方程，它刻画了对 N 和 D 的同时依赖关系，并支配过拟合的程度：

$$L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_D} \quad (1.5)$$

EN with fits pictured on the left in figure 4.

中 其拟合结果如图 4 左侧所示。

EN We conjecture that this functional form may also parameterize the trained log-likelihood for other generative modeling tasks.

中 我们推测，这一函数形式也可用于参数化其他生成式建模任务训练后的对数似然。

EN When training a given model for a finite number of parameter update steps S in the infinite data limit, after an initial transient period, the learning curves can be accurately fit by (see the right of figure 4)

中 在无限数据极限下，用有限的参数更新步骤数 S 训练给定模型时，经过最初的瞬态阶段之后，可以用下式精确拟合学习曲线（见图 4 右侧）

$$L(N, S) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{S_c}{S_{\min}(S)}\right)^{\alpha_S} \quad (1.6)$$

EN where $S_c \approx 2.1 \times 10^3$ and $\alpha_S \approx 0.76$, and $S_{\min}(S)$ is the minimum possible number of optimization steps (parameter updates)

中 其中 $S_c \approx 2.1 \times 10^3$, $\alpha_S \approx 0.76$, 而 $S_{\min}(S)$ 是可能的最少优化步骤数（参数更新次数），

EN estimated using Equation (5.4).

中 其值使用方程 (5.4) 进行估计。

EN When training within a fixed compute budget C , but with no other constraints, Equation (1.6) leads to the prediction that the optimal model size N , optimal batch size B , optimal number of steps S , and dataset size D should grow as

中 当在固定计算预算 C 内进行训练、但没有其他约束时，方程 (1.6) 可以预测，最优模型规模 N 、最优 batch size B 、最优步骤数 S 和数据集大小 D 应按如下方式增长：

$$N \propto C^{\alpha_C^{\min}/\alpha_N}, \quad B \propto C^{\alpha_C^{\min}/\alpha_B}, \quad S \propto C^{\alpha_C^{\min}/\alpha_S}, \quad D = B \cdot S \quad (1.7)$$

EN with

中 其中

$$\alpha_C^{\min} = 1 / (1/\alpha_S + 1/\alpha_B + 1/\alpha_N) \quad (1.8)$$

EN which closely matches the empirically optimal results $N \propto C_{\min}^{0.73}$, $B \propto C_{\min}^{0.24}$, and $S \propto C_{\min}^{0.03}$.

中 这与经验最优结果 $N \propto C_{\min}^{0.73}$ 、 $B \propto C_{\min}^{0.24}$ 和 $S \propto C_{\min}^{0.03}$ 非常吻合。

EN As the computational budget C increases, it should be spent primarily on larger models, without dramatic increases in training time or dataset size (see Figure 3).

中 随着计算预算 C 增加，它应主要用于更大的模型，而无需大幅增加训练时间或数据集大小（见图 3）。

EN This also implies that as models grow larger, they become increasingly sample efficient.

中 这也意味着随着模型变大，它们的样本效率会越来越高。

EN In practice, researchers typically train smaller models for longer than would be maximally compute-efficient because of hardware constraints.

中 在实践中，由于硬件约束，研究人员通常会采用比计算效率最高的配置更小的模型，并训练更长时间。

EN Optimal performance depends on total compute as a power law (see Equation (1.3)).

中 最优性能以幂律方式取决于总计算量（见方程 (1.3)）。

EN We provide some basic theoretical motivation for Equation (1.5), an analysis of learning curve fits and their implications for training time, and a breakdown of our results per token.

中 我们为方程 (1.5) 提供了一些基本的理论依据，分析了学习曲线拟合及其对训练时间的影响，并按 token 细分了我们的结果。

EN We also make some brief comparisons to LSTMs and recurrent Transformers [DGV⁺18].

中 我们还将这些结果与 LSTM 和循环 Transformer [DGV⁺18] 进行了简要比较。

1.3 Notation

符号约定

EN We use the following notation:

中 我们使用以下符号：

- **EN** L – the cross entropy loss in nats.

中 L ——以自然信息单位（纳特）计的交叉熵损失。

EN Typically it will be averaged over the tokens in a context, but in some cases we report the loss for specific tokens within the context.

中 通常会对一个上下文中的 tokens 取平均值，但在某些情况下，我们会报告上下文中特定 tokens 的损失。

• EN N – the number of model parameters, *excluding all vocabulary and positional embeddings*

中 N ——模型参数数量，不包括所有 *vocabulary embeddings* 和 *positional embeddings*

• EN $C \approx 6NBS$ – an estimate of the total non-embedding training compute, where B is the batch size, and S is the number of training steps (ie parameter updates).

中 $C \approx 6NBS$ ——对 non-embedding 训练总计算量的估计，其中 B 是 batch size， S 是训练步骤数（即参数更新次数）。

EN We quote numerical values in PF-days, where one PF-day = $10^{15} \times 24 \times 3600 = 8.64 \times 10^{19}$ floating point operations.

中 我们以 PF-日为单位给出数值，其中一个 PF-日 = $10^{15} \times 24 \times 3600 = 8.64 \times 10^{19}$ 次浮点运算。

• EN D – the dataset size in tokens

中 D ——以 tokens 计的数据集大小

• EN B_{crit} – the critical batch size [MKAT18], defined and discussed in Section 5.1.

中 B_{crit} ——critical batch size [MKAT18]，其定义和讨论见第 5.1 节。

EN Training at the critical batch size provides a roughly optimal compromise between time and compute efficiency.

中 使用 critical batch size 进行训练，可以在时间效率与计算效率之间实现大致最优的折中。

• EN C_{min} – an estimate of the minimum amount of non-embedding compute to reach a given value of the loss.

中 C_{min} ——为达到给定损失值所需的最少 non-embedding 计算量的估计。

EN This is the training compute that would be used if the model were trained at a batch size much less than the critical batch size.

中 如果模型采用远小于 critical batch size 的 batch size 进行训练，这就是会使用的训练计算量。

• EN S_{min} – an estimate of the minimal number of training steps needed to reach a given value of the loss.

中 S_{min} ——为达到给定损失值所需的最少训练步骤数的估计。

EN This is also the number of training steps that would be used if the model were trained at a batch size much greater than the critical batch size.

中 这也是在模型采用远大于 critical batch size 的 batch size 进行训练时会使用的训练步骤数。

• EN α_X – power-law exponents for the scaling of the loss as $L(X) \propto 1/X^{\alpha_X}$ where X can be any of $N, D, C, S, B, C^{\text{min}}$.

中 α_X ——损失按 $L(X) \propto 1/X^{\alpha_X}$ 缩放时的幂律指数，其中 X 可以是 $N, D, C, S, B, C^{\text{min}}$ 中的任意一个。

2 Background and Methods

背景和方法

EN We train language models on WebText2, an extended version of the WebText [RWC⁺19] dataset, tokenized using byte-pair encoding [SHB15] with a vocabulary size $n_{\text{vocab}} = 50257$.

中 我们在 WebText [RWC⁺19] 数据集的扩展版本 WebText2 上训练语言模型；该数据集使用字节对编码 [SHB15] 进行 tokenization，词表大小为 $n_{\text{vocab}} = 50257$ 。

EN We optimize the autoregressive log-likelihood (i.e. cross-entropy loss) averaged over a 1024-token context, which is also our principal performance metric.

中 我们优化在包含 1024 个 tokens 的上下文上取平均值的 autoregressive 对数似然（即交叉熵损失），这也是我们的主要性能指标。

Operation	Parameters	FLOPs per Token
Embed	$(n_{\text{vocab}} + n_{\text{ctx}}) d_{\text{model}}$	$4d_{\text{model}}$
Attention: QKV	$n_{\text{layer}} d_{\text{model}} 3d_{\text{attn}}$	$2n_{\text{layer}} d_{\text{model}} 3d_{\text{attn}}$
Attention: Mask	—	$2n_{\text{layer}} n_{\text{ctx}} d_{\text{attn}}$
Attention: Project	$n_{\text{layer}} d_{\text{attn}} d_{\text{model}}$	$2n_{\text{layer}} d_{\text{attn}} d_{\text{embd}}$
Feedforward	$n_{\text{layer}} 2d_{\text{model}} d_{\text{ff}}$	$2n_{\text{layer}} 2d_{\text{model}} d_{\text{ff}}$
De-embed	—	$2d_{\text{model}} n_{\text{vocab}}$
Total (Non-Embedding)	$N = 2d_{\text{model}} n_{\text{layer}} (2d_{\text{attn}} + d_{\text{ff}})$	$C_{\text{forward}} = 2N + 2n_{\text{layer}} n_{\text{ctx}} d_{\text{attn}}$

Table / 表 1 *EN* Parameter counts and compute (forward pass) estimates for a Transformer model.

中 Transformer 模型的参数量和计算量（前向传播）估计。

EN Sub-leading terms such as nonlinearities, biases, and layer normalization are omitted.

中 省略了非线性、偏置和层归一化等次要项。

EN We record the loss on the WebText2 test distribution and on a selection of other text distributions.

中 我们记录模型在 WebText2 测试分布以及若干其他文本分布上的损失。

EN We primarily train decoder-only [LSP⁺18, RNSS18] Transformer [VSP⁺17] models, though we also train LSTM models and Universal Transformers [DGV⁺18] for comparison.

中 我们主要训练仅含解码器的 [LSP⁺18, RNSS18] Transformer [VSP⁺17] 模型，不过也训练了 LSTM 模型和通用 Transformer [DGV⁺18] 用于比较。

2.1 Parameter and Compute Scaling of Transformers

Transformer 参数量与计算量的缩放关系

EN We parameterize the Transformer architecture using hyperparameters n_{layer} (number of layers), d_{model} (dimension of the residual stream), d_{ff} (dimension of the intermediate feed-forward layer), d_{attn} (dimension of the attention output), and n_{heads} (number of attention heads per layer).

中 我们使用超参数 n_{layer} （层数）、 d_{model} （残差流的维度）、 d_{ff} （中间前馈层的维度）、 d_{attn} （注意力输出的维度）和 n_{heads} （每层的注意力头数）来参数化 Transformer 架构。

EN We include n_{ctx} tokens in the input context, with $n_{\text{ctx}} = 1024$ except where otherwise noted.

中 我们在输入上下文中包含 n_{ctx} 个 tokens；除非另有说明，否则 $n_{\text{ctx}} = 1024$ 。

EN We use N to denote the model size, which we define as the number of *non-embedding* parameters

中 我们用 N 表示模型规模，并将其定义为 *non-embedding* parameters 的数量

$$\begin{aligned}
 N &\approx 2d_{\text{model}} n_{\text{layer}} (2d_{\text{attn}} + d_{\text{ff}}) \\
 &= 12n_{\text{layer}} d_{\text{model}}^2 \quad \text{with the standard} \quad d_{\text{attn}} = d_{\text{ff}}/4 = d_{\text{model}}
 \end{aligned} \tag{2.1}$$

EN where we have excluded biases and other sub-leading terms.

中 其中我们排除了偏置和其他次要项。

EN Our models also have $n_{\text{vocab}} d_{\text{model}}$ parameters in an embedding matrix, and use $n_{\text{ctx}} d_{\text{model}}$ parameters for positional embeddings, but we do not include these when discussing the ‘model size’ N ; we will see that this produces significantly cleaner scaling laws.

中 我们的模型在 embedding 矩阵中还包含 $n_{\text{vocab}} d_{\text{model}}$ 个参数，并使用 $n_{\text{ctx}} d_{\text{model}}$ 个参数表示位置 embeddings，但在讨论“模型规模” N 时，我们不包括这些参数；我们将看到，这样做会得到明显更简洁的 scaling laws。

EN Evaluating a forward pass of the Transformer involves roughly

中 对 Transformer 执行一次前向传播大致涉及

$$C_{\text{forward}} \approx 2N + 2n_{\text{layer}} n_{\text{ctx}} d_{\text{model}} \tag{2.2}$$

EN add-multiply operations, where the factor of two comes from the multiply-accumulate operation used in matrix multiplication.

中 次加乘运算，其中系数 2 来自矩阵乘法所使用的乘加运算。

EN A more detailed per-operation parameter and compute count is included in Table 1.

中 表 1 中给出了按操作划分的更详细参数量和计算量。

EN For contexts and models with $d_{\text{model}} > n_{\text{ctx}}/12$, the context-dependent computational cost per token is a relatively small fraction of the total compute.

中 对于满足 $d_{\text{model}} > n_{\text{ctx}}/12$ 的上下文和模型，每个 token 中依赖上下文的计算成本只占总计算量中相对较小的一部分。

EN Since we primarily study models where $d_{\text{model}} \gg n_{\text{ctx}}/12$, we do not include context-dependent terms in our training compute estimate.

中 由于我们主要研究 $d_{\text{model}} \gg n_{\text{ctx}}/12$ 的模型，因此在估算训练计算量时不包括依赖上下文的项。

EN Accounting for the backwards pass (approximately twice the compute as the forwards pass), we then define the estimated non-embedding compute as $C \approx 6N$ floating point operators per training token.

中 计入反向传播（其计算量约为前向传播的两倍）后，我们将估计的 non-embedding 计算量定义为每个训练 token $C \approx 6N$ 次浮点运算。

2.2 Training Procedures

训练流程

EN Unless otherwise noted, we train models with the Adam optimizer [KB14] for a fixed 2.5×10^5 steps with a batch size of 512 sequences of 1024 tokens.

中 除非另有说明，否则我们使用 Adam 优化器 [KB14] 对模型进行固定 2.5×10^5 步的训练，batch size 为 512 个序列，每个序列包含 1024 个 tokens。

EN Due to memory constraints, our largest models (more than 1B parameters) were trained with Adafactor [SS18].

中 由于内存限制，我们最大的模型（超过 10 亿个参数）使用 Adafactor [SS18] 进行训练。

EN We experimented with a variety of learning rates and schedules, as discussed in Appendix D.6.

中 我们试验了多种 learning rate 和调度方案，相关讨论见附录 D.6。

EN We found that results at convergence were largely independent of learning rate schedule.

中 我们发现，收敛时的结果基本不依赖于 learning-rate schedule。

EN Unless otherwise noted, all training runs included in our data used a learning rate schedule with a 3000 step linear warmup followed by a cosine decay to zero.

中 除非另有说明，否则数据中包含的所有训练过程都采用一种 learning-rate schedule 方案：先进行 3000 步线性预热，随后按余弦方式衰减至零。

2.3 Datasets

数据集

EN We train our models on an extended version of the WebText dataset described in [RWC⁺19].

中 我们在 [RWC⁺19] 所述 WebText 数据集的扩展版本上训练模型。

EN The original WebText dataset was a web scrape of outbound links from Reddit through December 2017 which received at least 3 karma.

中 原始 WebText 数据集通过抓取截至 2017 年 12 月、获得至少 3 点社区积分的 Reddit 外链而得到。

EN In the second version, WebText2, we added outbound Reddit links from the period of January to October 2018, also with a minimum of 3 karma.

中 在第二个版本 WebText2 中，我们加入了 2018 年 1 月至 10 月期间的 Reddit 外链，同样要求社区积分至少为 3 点。

EN The karma threshold served as a heuristic for whether people found the link interesting or useful.

中 社区积分阈值被用作判断人们是否认为链接有趣或有用的启发式指标。

EN The text of the new links was extracted with the Newspaper3k python library.

中 新链接中的文本是使用 Newspaper3k Python 库提取的。

EN In total, the dataset consists of 20.3M documents containing 96 GB of text and 1.62×10^{10} words (as defined by `wc`).

中 整个数据集共包含 2030 万篇文档、96 GB 文本和 1.62×10^{10} 个单词（按 `wc` 的定义）。

EN We then apply the reversible tokenizer described in [RWC⁺19], which yields 2.29×10^{10} tokens.

中 随后，我们应用 [RWC⁺19] 中所述的可逆 tokenizer，得到 2.29×10^{10} 个 tokens。

EN We reserve 6.6×10^8 of these tokens for use as a test set, and we also test on similarly-prepared samples of Books Corpus [ZKZ⁺15], Common Crawl [Fou], English Wikipedia, and a collection of publicly-available Internet Books.

中 我们留出其中的 6.6×10^8 个 tokens 用作测试集，并且还在以类似方式制备的图书语料库 [ZKZ⁺15]、公共网页抓取语料库 [Fou]、英文维基百科和公开可用的互联网图书集合样本上进行测试。

3 Empirical Results and Basic Power Laws

经验结果和基本幂律

EN To characterize language model scaling we train a wide variety of models, varying a number of factors including:

中 为了刻画语言模型的缩放特性，我们训练了多种多样的模型，并改变了若干因素，其中包括：

- **EN** Model size (ranging in size from 768 to 1.5 billion non-embedding parameters)
 - 中 模型规模（从 768 个到 15 亿个 non-embedding parameters）
- **EN** Dataset size (ranging from 22 million to 23 billion tokens)
 - 中 数据集大小（从 2200 万到 230 亿个 tokens）
- **EN** Shape (including depth, width, attention heads, and feed-forward dimension)
 - 中 形状（包括深度、宽度、注意力头和前馈维度）
- **EN** Context length (1024 for most runs, though we also experiment with shorter contexts)
 - 中 上下文长度（大多数训练过程为 1024，不过我们也试验了更短的上下文）
- **EN** Batch size (2^{19} for most runs, but we also vary it to measure the critical batch size)
 - 中 batch size（大多数训练过程为 2^{19} ，但我们也通过改变它来测量 critical batch size）

EN In this section we will display data along with empirically-motivated fits, deferring theoretical analysis to later sections.

中 本节将展示数据和受经验启发的拟合，并将理论分析留至后面的章节。

3.1 Approximate Transformer Shape and Hyperparameter Independence

Transformer 形状和超参数的近似独立性

EN Transformer performance depends very weakly on the shape parameters n_{layer} , n_{heads} , and d_{ff} when we hold the total non-embedding parameter count N fixed.

中 在我们保持 non-embedding parameter count N 不变时，Transformer 性能对形状参数 n_{layer} , n_{heads} 和 d_{ff} 的依赖非常弱。

EN To establish these results we trained models with fixed size while varying a single hyperparameter.

中 为了得到这些结果，我们在固定模型规模的同时改变单个超参数来训练模型。

EN This was simplest for the case of n_{heads} .

中 对于 n_{heads} ，这一操作最为简单。

EN When varying n_{layer} , we simultaneously varied d_{model} while keeping $N \approx 12n_{\text{layer}}d_{\text{model}}^2$ fixed.

中 在改变 n_{layer} 时，我们同时改变 d_{model} ，同时保持 $N \approx 12n_{\text{layer}}d_{\text{model}}^2$ 不变。

EN Similarly, to vary d_{ff} at fixed model size we also simultaneously varied the d_{model} parameter, as required by the parameter counts in Table 1.

中 同样，为了在固定模型规模的情况下改变 d_{ff} ，我们也按表 1 中参数量所要求的方式同时改变 d_{model} 参数。

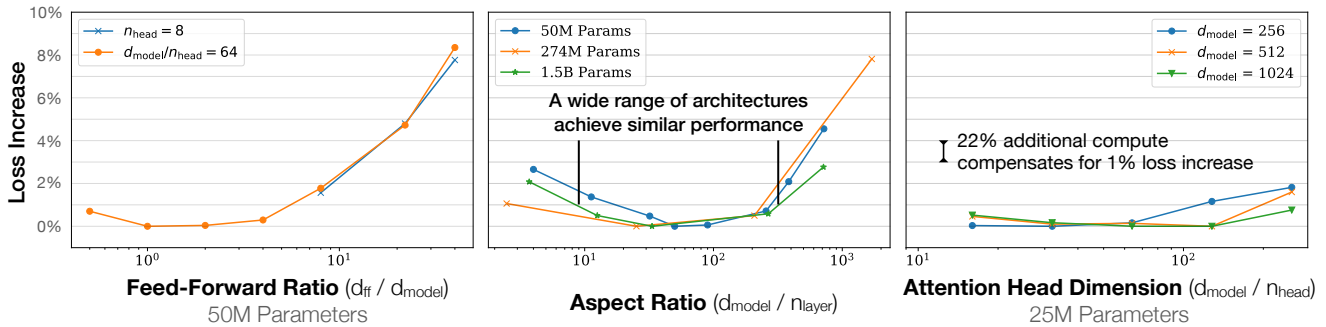


Figure / 图 5 EN Performance depends very mildly on model shape when the total number of non-embedding parameters N is held fixed.

中 当 non-embedding parameters 总数 N 保持不变时，性能对模型形状的依赖非常弱。

EN The loss varies only a few percent over a wide range of shapes.

中 在很大的形状范围内，损失的变化只有百分之几。

EN Small differences in parameter counts are compensated for by using the fit to $L(N)$ as a baseline.

中 参数数量的微小差异通过使用 $L(N)$ 的拟合作为基线来补偿。

EN Aspect ratio in particular can vary by a factor of 40 while only slightly impacting performance; an $(n_{\text{layer}}, d_{\text{model}}) = (6, 4288)$ reaches a loss within 3% of the $(48, 1600)$ model used in [RWC⁺19].

中 尤其是，纵横比即使相差 40 倍，对性能的影响也很小； $(n_{\text{layer}}, d_{\text{model}}) = (6, 4288)$ 所达到的损失与 [RWC⁺19] 中使用的 $(48, 1600)$ 模型相差不到 3%。

EN Independence of n_{layers} would follow if deeper Transformers effectively behave as ensembles of shallower models, as has been suggested for ResNets [VWB16].

中 如果更深的 Transformer 实际上表现得像较浅模型的集成，就会得到 n_{layers} 的独立性，正如针对 ResNet 所提出的观点 [VWB16]。

EN The results are shown in Figure 5.

中 结果如图 5 所示。

3.2 Performance with Non-Embedding Parameter Count N 性能与 non-embedding parameter count N 的关系

EN In Figure 6 we display the performance of a wide variety of models, ranging from small models with shape $(n_{\text{layer}}, d_{\text{model}}) = (2, 128)$ through billion-parameter models, ranging in shape from $(6, 4288)$ through $(207, 768)$.

中 在图 6 中，我们展示了众多不同模型的性能，范围从形状为 $(n_{\text{layer}}, d_{\text{model}}) = (2, 128)$ 的小模型，一直到形状介于 $(6, 4288)$ 和 $(207, 768)$ 之间的十亿参数模型。

EN Here we have trained to near convergence on the full WebText2 dataset and observe no overfitting (except possibly for the very largest models).

中 这里，我们在完整的 WebText2 数据集上将模型训练到接近收敛，并且没有观察到过拟合（最大的模型可能除外）。

EN As shown in Figure 1, we find a steady trend with non-embedding parameter count N , which can be fit to the first term of Equation (1.5), so that

中 如图 1 所示，我们发现 non-embedding parameter count N 呈现稳定趋势，可以用方程 (1.5) 的第一项拟合，即

$$L(N) \approx \left(\frac{N_c}{N} \right)^{\alpha_N} \quad (3.1)$$

EN To observe these trends it is crucial to study performance as a function of N ; if we instead use the total parameter count (including the embedding parameters) the trend is somewhat obscured (see Figure 6).

中 要观察到这些趋势，将性能视为 N 的函数进行研究至关重要；如果改用总参数数量（包括 embedding parameters），这一趋势就会在一定程度上变得模糊（见图 6）。

EN This suggests that the embedding matrix can be made smaller without impacting performance, as has been seen in recent work [LCG⁺19].

中 这表明可以在不影响性能的情况下缩小 embedding 矩阵，近期研究 [LCG⁺19] 中也观察到了这一点。

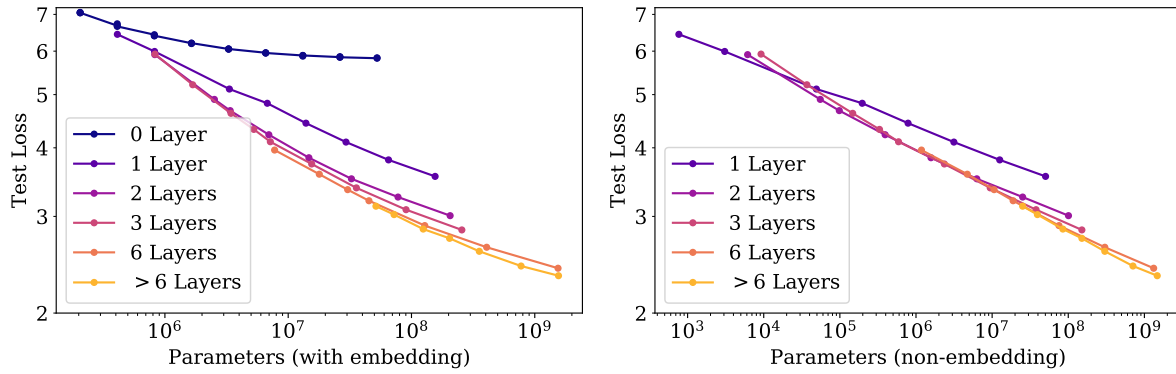


Figure / 图 6 *EN* Left: When we include embedding parameters, performance appears to depend strongly on the number of layers in addition to the number of parameters.

中 左图：当我们包括 embedding parameters 时，除了参数数量之外，性能似乎还强烈依赖于层数。

EN Right: When we exclude embedding parameters, the performance of models with different depths converge to a single trend.

中 右图：当我们排除 embedding parameters 时，不同深度模型的性能会收敛到同一趋势。

EN Only models with fewer than 2 layers or with extreme depth-to-width ratios deviate significantly from the trend.

中 只有层数少于 2 层或深宽比极端的模型才会显著偏离这一趋势。

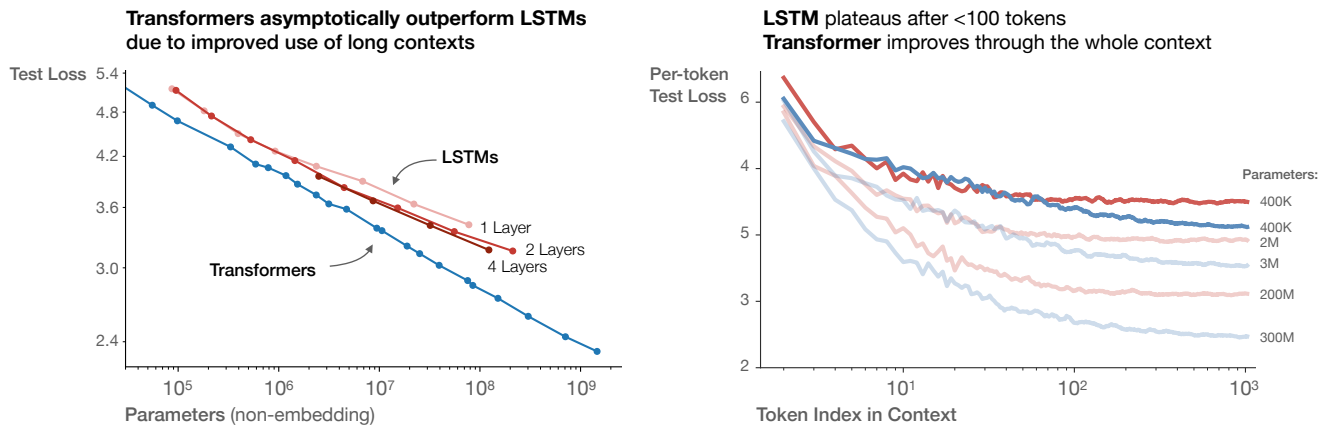


Figure / 图 7 *EN* LSTM and Transformer performance comparison

中 LSTM 与 Transformer 的性能比较

EN Although these models have been trained on the WebText2 dataset, their test loss on a variety of other datasets is also a power-law in N with nearly identical power, as shown in Figure 8.

中 尽管这些模型是在 WebText2 数据集上训练的，但如图 8 所示，它们在其他多种数据集上的测试损失也与 N 呈幂律关系，并且幂指数几乎相同。

3.2.1 Comparing to LSTMs and Universal Transformers 与 LSTM 和通用 Transformer 的比较

EN In Figure 7 we compare LSTM and Transformer performance as a function of non-embedding parameter count N .

中 在图 7 中，我们比较了 LSTM 和 Transformer 的性能随 non-embedding parameter count N 的变化。

EN The LSTMs were trained with the same dataset and context length.

中 LSTM 使用相同的数据集和上下文长度进行训练。

EN We see from these figures that the LSTMs perform as well as Transformers for tokens appearing early in the context, but cannot match the Transformer performance for later tokens.

中 从这些图中可以看到，对于出现在上下文前部的 tokens，LSTM 的表现与 Transformer 一样好，但对于后部的 tokens，它们无法达到 Transformer 的性能。

EN We present power-law relationships between performance and context position Appendix D.5, where increasingly large powers for larger models suggest improved ability to quickly recognize patterns.

中 在附录 D.5 中，我们给出了性能与上下文位置之间的幂律关系；对于更大的模型，幂指数越来越大，这表明其快速识别模式的能力得到提升。

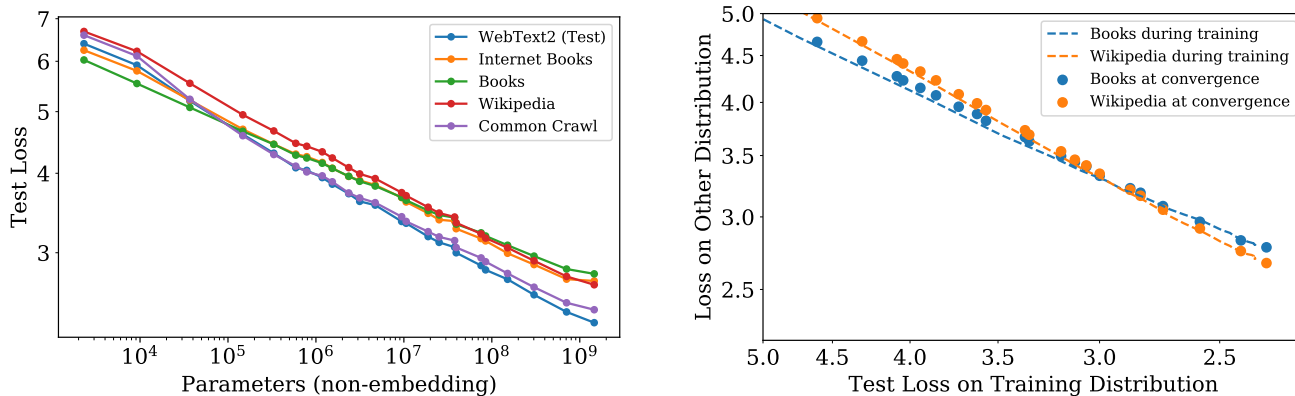


Figure / 图 8 **EN Left:** Generalization performance to other data distributions improves smoothly with model size, with only a small and very slowly growing offset from the WebText2 training distribution.

中 左图：对其他数据分布的泛化性能随模型规模平滑提升，与 WebText2 训练分布之间只有很小且增长极其缓慢的偏移。

EN Right: Generalization performance depends only on training distribution performance, and not on the phase of training.

中 右图：泛化性能只取决于训练分布上的性能，而与训练阶段无关。

EN We compare generalization of converged models (points) to that of a single large model (dashed curves) as it trains.

中 我们将已收敛模型的泛化性能（点）与单个大模型在训练过程中的泛化性能（虚线曲线）进行比较。

EN We also compare the performance of standard Transformers to recurrent Transformers [DGV⁺18] in Figure 17 in the appendix.

中 我们还比较了标准 Transformer 与循环 Transformer [DGV⁺18] 的性能，结果见附录中的图 17。

EN These models re-use parameters, and so perform slightly better as a function of N , at the cost of additional compute per-parameter.

中 这些模型会复用参数，因此作为 N 的函数，其性能略好，但代价是每个参数需要额外的计算量。

3.2.2 Generalization Among Data Distributions 数据分布之间的泛化

EN We have also tested our models on a set of additional text data distributions.

中 我们还在一组额外的文本数据分布上测试了模型。

EN The test loss on these datasets as a function of model size is shown in Figure 8; in all cases the models were trained only on the WebText2 dataset.

中 这些数据集上的测试损失随模型规模的变化如图 8 所示；在所有情况下，模型都只在 WebText2 数据集上进行了训练。

EN We see that the loss on these other data distributions improves smoothly with model size, in direct parallel with the improvement on WebText2.

中 可以看到，其他数据分布上的损失随模型规模平滑改善，与 WebText2 上的改善完全同步。

EN We find that generalization depends almost exclusively on the in-distribution validation loss, and does not depend on the duration of training or proximity to convergence.

中 我们发现，泛化几乎只取决于分布内验证损失，而不取决于训练持续时间或接近收敛的程度。

EN We also observe no dependence on model depth (see Appendix D.8).

中 我们也没有观察到它对模型深度的依赖（见附录 D.8）。

3.3 Performance with Dataset Size and Compute

性能与数据集大小及计算量的关系

EN We display empirical trends for the test loss as a function of dataset size D (in tokens) and training compute C in Figure 1.

中 我们在图 1 中展示了测试损失随数据集大小 D （以 tokens 计）和训练计算量 C 变化的经验趋势。

EN For the trend with D we trained a model with $(n_{\text{layer}}, n_{\text{embd}}) = (36, 1280)$ on fixed subsets of the WebText2 dataset.

中 为了研究随 D 变化的趋势，我们在 WebText2 数据集的固定子集上训练了一个 $(n_{\text{layer}}, n_{\text{embd}}) = (36, 1280)$ 的模型。

EN We stopped training once the test loss ceased to decrease.

中 一旦测试损失不再下降，我们就停止训练。

EN We see that the resulting test losses can be fit with simple power-law

中 可以看到，由此得到的测试损失可以用关于数据集大小的简单幂律

$$L(D) \approx \left(\frac{D_c}{D}\right)^{\alpha_D} \quad (3.2)$$

EN in the dataset size.

中 进行拟合。

EN The data and fit appear in Figure 1.

中 数据和拟合结果如图 1 所示。

EN The total amount of non-embedding compute used during training can be estimated as $C = 6NBS$, where B is the batch size, S is the number of parameter updates, and the factor of 6 accounts for the forward and backward passes.

中 训练期间使用的 non-embedding 计算总量可以估计为 $C = 6NBS$ ，其中 B 是 batch size， S 是参数更新次数，而系数 6 用于计入前向传播和反向传播。

EN Thus for a given value of C we can scan over all models with various N to find the model with the best performance on step $S = \frac{C}{6BS}$.

中 因此，对于给定的 C 值，我们可以扫描具有不同 N 的所有模型，找出在第 $S = \frac{C}{6BS}$ 步上性能最好的模型。

EN Note that in these results *the batch size B remains fixed for all models*, which means that these empirical results are not truly optimal.

中 请注意，在这些结果中，所有模型的 *batch size B* 都保持不变，这意味着这些经验结果并非真正最优。

EN We will account for this in later sections using an adjusted $C_{\min}$ to produce cleaner trends.

中 我们将在后面的章节中使用经过调整的 $C_{\min}$ 来考虑这一点，从而得到更清晰的趋势。

EN The result appears as the heavy black line on the left-hand plot in Figure 1.

中 结果显示为图 1 左图中的粗黑线。

EN It can be fit with

中 它可以用下式拟合：

$$L(C) \approx \left(\frac{C_c}{C}\right)^{\alpha_C} \quad (3.3)$$

EN The figure also includes images of individual learning curves to clarify when individual models are optimal.

中 该图还展示了各个模型的学习曲线，用以阐明它们分别在何时最优。

EN We will study the optimal allocation of compute more closely later on.

中 我们将在后文更深入地研究计算量的最优分配。

EN The data strongly suggests that sample efficiency improves with model size, and we also illustrate this directly in Figure 19 in the appendix.

中 数据强烈表明，样本效率随模型规模增大而提高；我们还在附录的图 19 中直接展示了这一点。

4 Charting the Infinite Data Limit and Overfitting

描绘无限数据极限与过拟合

EN In Section 3 we found a number of basic scaling laws for language modeling performance.

中 在第 3 节中，我们发现了语言建模性能的若干基本 scaling laws。

EN Here we will study the performance of a model of size N trained on a dataset with D tokens while varying N and D simultaneously.

中 这里，我们将研究规模为 N 、在含有 D 个 tokens 的数据集上训练的模型，并同时改变 N 和 D 。

EN We will empirically demonstrate that the optimally trained test loss accords with the scaling law of Equation (1.5).

中 我们将通过实验证明，经过最优训练的测试损失符合公式 (1.5) 的 scaling law。

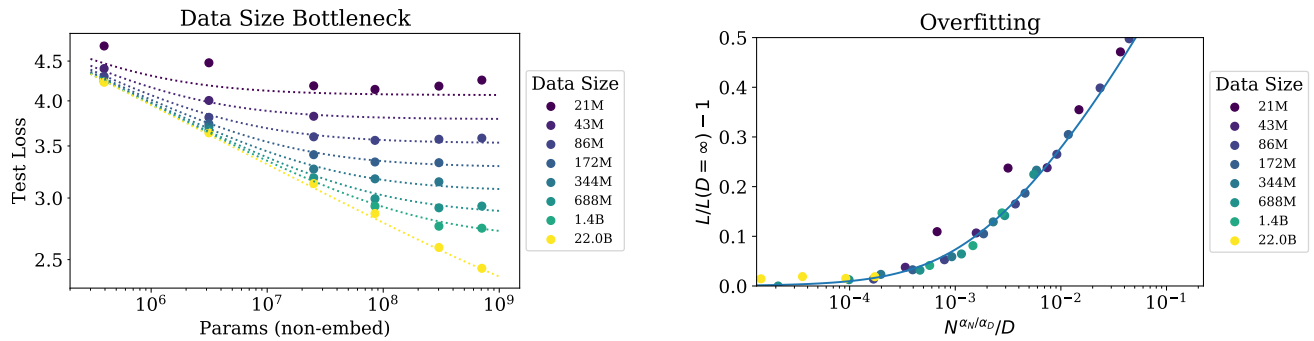


Figure / 图 9 **EN** The early-stopped test loss $L(N, D)$ depends predictably on the dataset size D and model size N according to Equation (1.5).

中 根据公式 (1.5)，early-stopped 测试损失 $L(N, D)$ 以可预测的方式取决于数据集大小 D 和模型规模 N 。

EN Left: For large D , performance is a straight power law in N .

中 左图：当 D 很大时，性能关于 N 呈严格的幂律关系。

EN For a smaller fixed D , performance stops improving as N increases and the model begins to overfit.

中 当固定的 D 较小时，随着 N 增大，性能会停止改善，模型开始过拟合。

EN (The reverse is also true, see Figure 4.)

中 （反过来也成立，见图 4。）

EN Right: The extent of overfitting depends predominantly on the ratio $N^{\frac{\alpha_N}{\alpha_D}}/D$, as predicted in equation (4.3).

中 右图：正如公式 (4.3) 所预测的，过拟合的程度主要取决于比率 $N^{\frac{\alpha_N}{\alpha_D}}/D$ 。

EN The line is our fit to that equation.

中 曲线是我们对该公式的拟合。

EN This provides guidance on how much data we would need to train models of increasing size while keeping overfitting under control.

中 这为我们提供了指导：在控制过拟合的同时，训练规模不断增大的模型需要多少数据。

4.1 **Proposed $L(N, D)$ Equation** 提议的 $L(N, D)$ 公式

EN We have chosen the parameterization (1.5) (repeated here for convenience):

中 我们选择了参数化形式 (1.5)（为方便起见在此重列）：

$$L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_D} \quad (4.1)$$

EN using three principles:

中 这基于三个原则：

1. **EN** Changes in vocabulary size or tokenization are expected to rescale the loss by an overall factor.

中 词表大小或 tokenization 的变化预计会使损失按一个整体因子重新缩放。

EN The parameterization of $L(N, D)$ (and all models of the loss) must naturally allow for such a rescaling.

中 $L(N, D)$ 的参数化形式（以及损失的所有模型）必须自然地允许这种重新缩放。

2. **EN** Fixing D and sending $N \rightarrow \infty$, the overall loss should approach $L(D)$.

中 固定 D 并令 $N \rightarrow \infty$ ，整体损失应趋近于 $L(D)$ 。

EN Conversely, fixing N and sending $D \rightarrow \infty$ the loss must approach $L(N)$.

中 反之，固定 N 并令 $D \rightarrow \infty$ ，损失必须趋近于 $L(N)$ 。

3. **EN** $L(N, D)$ should be analytic at $D = \infty$, so that it has a series expansion in $1/D$ with integer powers.

中 $L(N, D)$ 在 $D = \infty$ 处应当是解析的，从而它能够按 $1/D$ 的整数次幂作级数展开。

EN Theoretical support for this principle is significantly weaker than for the first two.

中 支持这一原则的理论依据明显弱于前两个原则。

EN Our choice of $L(N, D)$ satisfies the first requirement because we can rescale N_c, D_c with changes in the vocabulary.

中 我们选择的 $L(N, D)$ 满足第一项要求，因为词表变化时，我们可以重新缩放 N_c, D_c 。

EN This also implies that the values of N_c, D_c have no fundamental meaning.

中 这也意味着 N_c, D_c 的取值没有根本性的意义。

EN Since we stop training early when the test loss ceases to improve and optimize all models in the same way, we expect that larger models should always perform better than smaller models.

中 由于我们会在测试损失停止改善时进行 early stopping，并以相同方式优化所有模型，因此我们预计更大的模型应当始终优于更小的模型。

EN But with fixed finite D , we also do not expect any model to be capable of approaching the best possible loss (ie the entropy of text).

中 但在固定且有限的 D 下，我们也不认为任何模型能够接近可能达到的最佳损失（即文本的熵）。

EN Similarly, a model with fixed size will be capacity-limited.

中 类似地，规模固定的模型将受到容量限制。

EN These considerations motivate our second principle.

中 这些考虑促成了我们的第二个原则。

EN Note that knowledge of $L(N)$ at infinite D and $L(D)$ at infinite N fully determines all the parameters in $L(N, D)$.

中 请注意，知道无限 D 时的 $L(N)$ 和无限 N 时的 $L(D)$ ，就能完全确定 $L(N, D)$ 中的所有参数。

EN The third principle is more speculative.

中 第三个原则更具推测性。

EN There is a simple and general reason one might expect overfitting to scale $\propto 1/D$ at very large D .

中 我们可能预期，在 D 非常大时，过拟合按 $\propto 1/D$ 缩放，这背后有一个简单而普遍的原因。

EN Overfitting should be related to the variance or the signal-to-noise ratio of the dataset [AS17], and this scales as $1/D$.

中 过拟合应与数据集的方差或信噪比有关 [AS17]，而这会按 $1/D$ 缩放。

EN This expectation should hold for any smooth loss function, since we expect to be able to expand the loss about the $D \rightarrow \infty$ limit.

中 对于任何平滑的损失函数，这一预期都应成立，因为我们预计可以围绕 $D \rightarrow \infty$ 极限展开损失。

EN However, this argument assumes that $1/D$ corrections dominate over other sources of variance, such as the finite batch size and other limits on the efficacy of optimization.

中 然而，这一论证假设 $1/D$ 修正相较其他方差来源占主导地位，例如有限的 batch size 以及其他对优化效力的限制。

EN Without empirical confirmation, we would not be very confident of its applicability.

中 若无实验确认，我们不会对它的适用性抱有很强的信心。

EN Our third principle explains the asymmetry between the roles of N and D in Equation (1.5).

中 我们的第三个原则解释了公式 (1.5) 中 N 和 D 所扮演角色的不对称性。

EN Very similar symmetric expressions⁴ are possible, but they would not have a $1/D$ expansion with integer powers, and would require the introduction of an additional parameter.

中 非常相似的对称表达式⁵也是可能的，但它们无法按 $1/D$ 的整数次幂展开，并且还需要引入一个额外参数。

EN In any case, we will see that our equation for $L(N, D)$ fits the data well, which is the most important justification for our $L(N, D)$ ansatz.

中 无论如何，我们将会看到，关于 $L(N, D)$ 的公式能够很好地拟合数据；这正是我们的 $L(N, D)$ 假设形式最重要的依据。

4.2 Results

结果

⁴For example, one might have used $L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\alpha_N} + \left(\frac{D_c}{D} \right)^{\alpha_D} \right]^{\beta}$, but this does not have a $1/D$ expansion.

⁵例如，可以使用 $L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\alpha_N} + \left(\frac{D_c}{D} \right)^{\alpha_D} \right]^{\beta}$ ，但它不存在关于 $1/D$ 的展开。

EN We regularize all our models with 10% dropout, and by tracking test loss and stopping once it is no longer decreasing.

中 我们对所有模型都使用 10% 的 dropout 进行正则化，同时跟踪测试损失，并在其不再下降时停止训练。

EN The results are displayed in Figure 9, including a fit to the four parameters $\alpha_N, \alpha_D, N_c, D_c$ in Equation (1.5):

中 结果展示在图 9 中，其中还包括对公式 (1.5) 中四个参数 $\alpha_N, \alpha_D, N_c, D_c$ 的拟合：

Parameter	α_N	α_D	N_c	D_c
Value	0.076	0.103	6.4×10^{13}	1.8×10^{13}

Table / 表 2 **EN** Fits to $L(N, D)$

中 对 $L(N, D)$ 的拟合

EN We obtain an excellent fit, with the exception of the runs where the dataset has been reduced by a factor of 1024, to about 2×10^7 tokens.

中 除了数据集缩小了 1024 倍、降至约 2×10^7 个 tokens 的训练运行之外，我们得到了非常出色的拟合。

EN With such a small dataset, an epoch consists of only 40 parameter updates.

中 对于如此小的数据集，一个轮次仅包含 40 次参数更新。

EN Perhaps such a tiny dataset represents a different regime for language modeling, as overfitting happens very early in training (see Figure 16).

中 也许这样一个极小的数据集代表了语言建模的另一种状态，因为过拟合在训练的极早期就会发生（见图 16）。

EN Also note that the parameters differ very slightly from those obtained in Section 3, as here we are fitting the full $L(N, D)$ rather than just $L(N, \infty)$ or $L(\infty, D)$.

中 还请注意，这些参数与第 3 节中得到的参数略有不同，因为这里我们拟合的是完整的 $L(N, D)$ ，而不只是 $L(N, \infty)$ 或 $L(\infty, D)$ 。

EN To chart the borderlands of the infinite data limit, we can directly study the extent of overfitting.

中 为了描绘无限数据极限的边界地带，我们可以直接研究过拟合的程度。

EN For all but the largest models, we see no sign of overfitting when training with the full 22B token WebText2 dataset, so we can take it as representative of $D = \infty$.

中 除最大的模型之外，对于所有其他模型，使用完整的、含 22B tokens 的 WebText2 数据集训练时，我们都没有看到过拟合的迹象，因此可以将其视作 $D = \infty$ 的代表。

EN Thus we can compare finite D to the infinite data limit by defining

中 于是，我们可以通过定义

$$\delta L(N, D) \equiv \frac{L(N, D)}{L(N, \infty)} - 1 \quad (4.2)$$

EN and studying it as a function of N, D .

中 并将其作为 N, D 的函数来研究，从而比较有限 D 与无限数据极限。

EN In fact, we see empirically that δL depends only a specific combination of N and D , as shown in Figure 16.

中 事实上，如图 16 所示，我们从实验中看到， δL 仅取决于 N 和 D 的一个特定组合。

EN This follows from the scaling law of Equation (1.5), which implies

中 这可以由公式 (1.5) 的 scaling law 推出，该定律意味着

$$\delta L \approx \left(1 + \left(\frac{N}{N_c} \right)^{\frac{\alpha_N}{\alpha_D}} \frac{D_c}{D} \right)^{\alpha_D} - 1 \quad (4.3)$$

EN Note that at large D this formula also has a series expansion in powers of $1/D$.

中 请注意，在 D 很大时，该公式同样可以按 $1/D$ 的幂作级数展开。

EN We estimate that the variation in the loss with different random seeds is roughly 0.02, which means that to avoid overfitting when training to within that threshold of convergence we require

中 我们估计，在使用不同随机种子时，损失的变化约为 0.02；这意味着，如果希望训练至距离收敛不超过这一阈值时仍能避免过拟合，我们需要

$$D \gtrsim (5 \times 10^3) N^{0.74} \quad (4.4)$$

EN With this relation, models smaller than 10^9 parameters can be trained with minimal overfitting on the 22B token WebText2 dataset, but our largest models will encounter some mild overfitting.

中 按照这一关系，参数量少于 10^9 的模型可以在含 22B tokens 的 WebText2 数据集上训练，并且过拟合程度极小；但我们最大的模型将会遇到一些轻微的过拟合。

EN More generally, this relation shows that dataset size may grow sub-linearly in model size while avoiding overfitting.

中 更一般而言，这一关系表明，在避免过拟合的同时，数据集大小可以相对于模型规模作次线性增长。

EN Note however that this does not typically represent maximally compute-efficient training.

中 但请注意，这通常并不代表计算效率最高的训练。

EN We should also emphasize that we have not optimized regularization (eg the dropout probability) while varying dataset and model size.

中 我们还应强调，在改变数据集大小和模型规模时，我们并未优化正则化设置（例如 dropout 概率）。

5 Scaling Laws with Model Size and Training Time

模型规模与训练时间的 *Scaling Laws*

EN In this section we will demonstrate that a simple scaling law provides a good description for the loss as a function of model size N and training time.

中 在本节中，我们将证明，一条简单的 scaling law 能够很好地描述损失如何随模型规模 N 和训练时间变化。

EN First we will explain how to use the results of [MKAT18] to define a universal training step $S_{\min}$, which accounts for the fact that most of our models have not been trained at an optimal batch size.

中 首先，我们将说明如何利用 [MKAT18] 的结果来定义通用训练步数 $S_{\min}$ ，它考虑到了我们的大多数模型并未在最优 batch size 下训练这一事实。

EN Then we will demonstrate that we can fit the model size and training time dependence of the loss using Equation (1.6).

中 随后，我们将证明，可以使用公式 (1.6) 来拟合损失对模型规模和训练时间的依赖关系。

EN Later we will use these results to predict the optimal allocation of training compute between model size and training time, and then confirm that prediction.

中 之后，我们将使用这些结果来预测如何在模型规模和训练时间之间最优地分配训练计算量，然后验证这一预测。

5.1 Adjustment for Training at $B_{\text{crit}}(L)$

针对在 $B_{\text{crit}}(L)$ 下训练的校正

EN A simple empirical theory for the batch size dependence of training was developed in [MKAT18] (see also [SLA⁺18, ZLN⁺19]).

中 [MKAT18] 提出了一种关于训练如何依赖 batch size 的简单经验理论（另见 [SLA⁺18, ZLN⁺19]）。

EN It was argued that there is a critical batch size B_{crit} for training; for B up to B_{crit} the batch size can be increased with very minimal degradation in compute-efficiency, whereas for $B > B_{\text{crit}}$ increases in B result in diminishing returns.

中 文中指出，训练存在一个 critical batch size B_{crit} ；当 B 不超过 B_{crit} 时，可以增大 batch size，而计算效率几乎不会降低；当 $B > B_{\text{crit}}$ 时，增大 B 则会产生收益递减。

EN It was also argued that the gradient noise scale provides a simple prediction for B_{crit} , and that neither depends directly on model size except through the value of the loss that has been attained.

中 文中还指出，gradient noise scale 可以对 B_{crit} 作出简单预测，而且二者都不直接依赖于模型规模，只会通过已经达到的损失值间接依赖于它。

EN These results can be used to predict how training time and compute will vary with the batch size.

中 可以利用这些结果预测训练时间和计算量将如何随 batch size 变化。

EN To utilize both training time and compute as effectively as possible, it is best to train with a batch size $B \approx B_{\text{crit}}$.

中 为了尽可能有效地利用训练时间和计算量，最好使用 batch size $B \approx B_{\text{crit}}$ 进行训练。

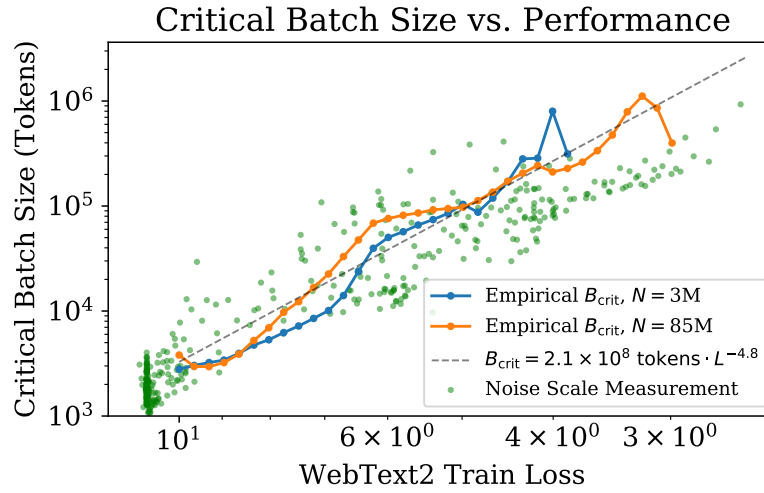


Figure / 图 10 *EN* The critical batch size B_{crit} follows a power law in the loss as performance increase, and does not depend directly on the model size.

中 随着性能提高，critical batch size B_{crit} 关于损失呈幂律关系，并且不直接依赖于模型规模。

EN We find that the critical batch size approximately doubles for every 13% decrease in loss.

中 我们发现，损失每降低 13%，critical batch size 大约翻一番。

EN B_{crit} is measured empirically from the data shown in Figure 18, but it is also roughly predicted by the gradient noise scale, as in [MKAT18].

中 B_{crit} 是根据图 18 所示数据通过实验测得的，但正如 [MKAT18] 中那样，gradient noise scale 也能对它作出大致预测。

EN Training at $B \gg B_{\text{crit}}$ minimizes the number of training steps, while $B \ll B_{\text{crit}}$ minimizes the use of compute.

中 在 $B \gg B_{\text{crit}}$ 时训练，会使训练步数最少；而在 $B \ll B_{\text{crit}}$ 时训练，则会使计算量最少。

EN More specifically, it was demonstrated that for a wide variety of neural network tasks, the number of training steps S and the number of data examples processed $E = BS$ satisfy the simple relation

中 更具体而言，已有研究证明，对于多种多样的神经网络任务，训练步数 S 和已处理的数据样本数 $E = BS$ 满足如下简单关系

$$\left(\frac{S}{S_{\min}} - 1\right) \left(\frac{E}{E_{\min}} - 1\right) = 1 \quad (5.1)$$

EN when training to any fixed value of the loss L .

中 该关系适用于训练达到任意固定损失值 L 的情形。

EN Here $S_{\min}$ is the minimum number of steps necessary to reach L , while $E_{\min}$ is the minimum number of data examples that must be processed.

中 这里， $S_{\min}$ 是达到 L 所需的最少步数，而 $E_{\min}$ 是必须处理的最少数据样本数。

EN We demonstrate the relation (5.1) for Transformers in Figure 18 in the appendix.

中 我们证明了关系 (5.1) 对 Transformer 同样成立，结果见附录中的图 18。

EN This relation defines the critical batch size

中 该关系定义了 critical batch size

$$B_{\text{crit}}(L) \equiv \frac{E_{\min}}{S_{\min}} \quad (5.2)$$

EN which is a function of the target value of the loss.

中 它是目标损失值的函数。

EN Training at the critical batch size makes a roughly optimal time/compute tradeoff, requiring $2S_{\min}$ training steps and processing $E = 2E_{\min}$ data examples.

中 在 critical batch size 下训练，可在时间与计算量之间作出大致最优的权衡，需要 $2S_{\min}$ 个训练步，并处理 $E = 2E_{\min}$ 个数据样本。

EN In Figure 10 we have plotted the critical batch size and gradient noise scale⁶ as a function of training loss for two different models.

中 在图 10 中，我们绘制了两个不同模型的 critical batch size 和 gradient noise scale⁷关于训练损失的函数。

EN We see that $B_{\text{crit}}(L)$ is independent of model size, and only depends on the loss L .

中 我们看到， $B_{\text{crit}}(L)$ 与模型规模无关，只取决于损失 L 。

EN So the predictions of [MKAT18] continue to hold for Transformer language models.

中 因此，[MKAT18] 的预测对 Transformer 语言模型仍然成立。

EN The critical batch size can be fit with a power-law in the loss

中 critical batch size 可以用关于损失的幂律来拟合：

$$B_{\text{crit}}(L) \approx \frac{B_*}{L^{1/\alpha_B}} \quad (5.3)$$

EN where $B_* \approx 2 \times 10^8$ and $\alpha_B \approx 0.21$.

中 其中 $B_* \approx 2 \times 10^8$ ， $\alpha_B \approx 0.21$ 。

EN We have chosen this parameterization for $B_{\text{crit}}(L)$ because as the loss approaches its minimum value $L_{\min}$, the gradient noise scale is expected to diverge, and we expect B_{crit} to track this noise scale.

中 我们之所以为 $B_{\text{crit}}(L)$ 选择这一参数化形式，是因为当损失趋近其最小值 $L_{\min}$ 时，gradient noise scale 预计会发散，而我们预计 B_{crit} 会追随这一 gradient noise scale。

EN We do not know $L_{\min}$, as we see no sign that our models are approaching it, but $L_{\min} > 0$ since the entropy of natural language is non-zero.

中 我们并不知道 $L_{\min}$ ，因为没有看到模型正在趋近它的迹象；但 $L_{\min} > 0$ ，因为自然语言的熵非零。

EN Since apparently $L_{\min}$ is much smaller than the values of L we have achieved, we used a parameterization where B_{crit} diverges as $L \rightarrow 0$.

中 由于 $L_{\min}$ 显然远小于我们已经达到的 L 值，因此我们使用了一种当 $L \rightarrow 0$ 时 B_{crit} 发散的参数化形式。

EN We will use $B_{\text{crit}}(L)$ to estimate the relation between the number of training steps S while training at batch size $B = 2^{19}$ tokens and the number of training steps while training at $B \gg B_{\text{crit}}$.

中 我们将使用 $B_{\text{crit}}(L)$ 来估计以下两者之间的关系：在 batch size $B = 2^{19}$ tokens 下训练时的训练步数 S ，以及在 $B \gg B_{\text{crit}}$ 下训练时的训练步数。

EN This is simply

中 这一关系就是

$$S_{\min}(S) \equiv \frac{S}{1 + B_{\text{crit}}(L)/B} \quad (\text{minimum steps, at } B \gg B_{\text{crit}}) \quad (5.4)$$

EN for any given target value L for the loss.

中 其中损失可以取任意给定的目标值 L 。

EN This also defines a critical value of the compute needed to train to L with a model of size N if we were to train at $B \ll B_{\text{crit}}(L)$.

中 如果我们在 $B \ll B_{\text{crit}}(L)$ 下训练，这也定义了使用规模为 N 的模型训练至 L 所需计算量的临界值。

EN This is

中 它是

$$C_{\min}(C) \equiv \frac{C}{1 + B/B_{\text{crit}}(L)} \quad (\text{minimum compute, at } B \ll B_{\text{crit}}) \quad (5.5)$$

EN where $C = 6NBS$ estimates the (non-embedding) compute used at batch size B .

中 其中 $C = 6NBS$ 估计了在 batch size B 下使用的 (non-embedding) 计算量。

5.2 Results for $L(N, S_{\min})$ and Performance with Model Size and Compute

$L(N, S_{\min})$ 的结果以及性能如何随模型规模和计算量变化

⁶Although the critical batch size roughly matches the gradient noise scale, we are using a direct measurements of B_{crit} from Figures 18 and 10 for all our later analyses.

⁷尽管 critical batch size 与 gradient noise scale 大致吻合，但在之后的所有分析中，我们使用的是根据图 18 和图 10 直接测得的 B_{crit} 。

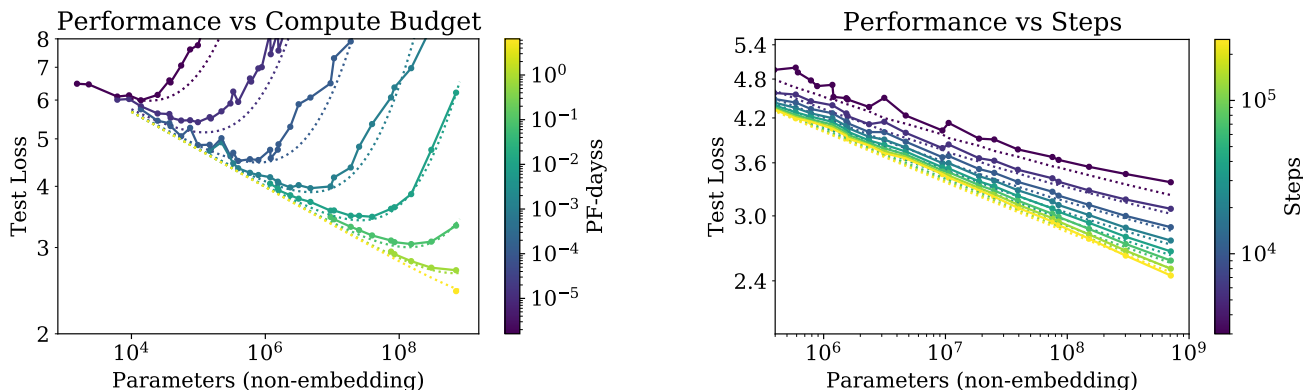


Figure / 图 11 *EN* When we hold either total compute or number of training steps fixed, performance follows $L(N, S)$ from Equation (5.6).

中 当我们固定总计算量或训练步数时，性能遵循公式 (5.6) 中的 $L(N, S)$ 。

EN Each value of compute budget has an associated optimal model size that maximizes performance.

中 每个计算预算值都有一个使性能最大化的相应最优模型规模。

EN Mediocre fits at small S are unsurprising, as the power-law equation for the learning curves breaks down very early in training.

中 在 S 较小时拟合效果一般并不令人意外，因为学习曲线的幂律公式会在训练极早期失效。

EN Now we will use $S_{\min}$ defined in Equation (5.4) to obtain a simple and universal fit for the dependence of the loss on model size and training time in the infinite data limit.

中 现在，我们将使用公式 (5.4) 中定义的 $S_{\min}$ ，对无限数据极限下损失对模型规模和训练时间的依赖关系进行简单而通用的拟合。

EN We will fit the stable, Adam-optimized training runs using Equation (1.6), repeated here for convenience:

中 我们将使用公式 (1.6)（为方便起见在此重列）拟合那些稳定且采用 Adam 优化的训练实验：

$$L(N, S_{\min}) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{S_c}{S_{\min}}\right)^{\alpha_S} \quad (5.6)$$

EN for the loss.

中 来表示损失。

EN We include all training steps after the warmup period of the learning rate schedule, and find a fit to the data with the parameters:

中 我们纳入了 learning-rate schedule 预热期之后的所有训练步，并发现数据可用以下参数拟合：

Parameter	α_N	α_S	N_c	S_c
Value	0.077	0.76	6.5×10^{13}	2.1×10^3

Table / 表 3 *EN* Fits to $L(N, S)$

中 对 $L(N, S)$ 的拟合

EN With these parameters, we obtain the learning curve fits in Figure 4.

中 利用这些参数，我们得到了图 4 中的学习曲线拟合。

EN Though the fits are imperfect, we believe they are quite compelling given the simplicity of Equation (5.6).

中 尽管拟合并不完美，但考虑到公式 (5.6) 如此简单，我们认为结果相当有说服力。

EN The data and fits can be visualized in a different and more interesting way, as shown in Figure 11.

中 如图 11 所示，还可以用一种不同且更有趣的方式将数据和拟合结果可视化。

EN There we study the test loss as a function of model size while fixing either the total non-embedding compute C used in training, or the number of steps S .

中 图中，我们研究了在固定训练所用的 non-embedding 总计算量 C 或步数 S 时，测试损失如何随模型规模变化。

EN For the fits we use Equation (5.5) and (5.4) along with the parameters above and Equation (5.6).

中 在拟合中，我们将公式 (5.5) 和 (5.4) 与上述参数以及公式 (5.6) 结合使用。

EN The power-law dependence of the loss on $S_{\min}$ reflects the interplay of optimizer dynamics and the loss landscape.

中 损失对 $S_{\min}$ 的幂律依赖反映了优化器动力学与损失景观之间的相互作用。

EN Since the fits are best late in training, when the loss may be approximately quadratic, the power-law should provide information about the spectrum of the Hessian of the loss.

中 由于在训练后期的拟合效果最好，而此时损失可能近似为二次函数，因此该幂律应当能够提供关于损失的 Hessian 矩阵谱的信息。

EN Its universality suggests that the Hessian eigenvalue density is roughly independent of model size.

中 它的普适性表明，Hessian 矩阵的特征值密度大致与模型规模无关。

5.3 Lower Bound on Early Stopping Step Early Stopping 步数的下界

EN The results for $L(N, S_{\min})$ can be used to derive a lower-bound (and rough estimate) of the step at which early stopping should occur when training is data limited.

中 $L(N, S_{\min})$ 的结果可用于推导在数据受限训练中应当进行 early stopping 的步数下界（以及粗略估计）。

EN It is motivated by the idea that finite and infinite D learning curves for a given model will be very similar until we reach $S_{\min} \approx S_{\text{stop}}$.

中 其动机是：对于给定模型，有限 D 与无限 D 时的学习曲线会非常相似，直到达到 $S_{\min} \approx S_{\text{stop}}$ 。

EN Thus overfitting should be proportional to the correction from simply ending training at S_{stop} .

中 因此，过拟合应当与仅仅在 S_{stop} 时结束训练所带来的修正成正比。

EN This will underestimate S_{stop} , because in reality the test loss will decrease more slowly when we have a finite D , and therefore we will require more training steps to reach the optimal test loss at finite D .

中 这将低估 S_{stop} ，因为现实中，在 D 有限时，测试损失会下降得更慢，因此需要更多训练步才能在有限 D 下达到最优测试损失。

EN This line of reasoning leads to the inequality

中 这一推理过程导出如下不等式：

$$S_{\text{stop}}(N, D) \gtrsim \frac{S_c}{[L(N, D) - L(N, \infty)]^{1/\alpha_S}} \quad (5.7)$$

EN where $L(N, \infty)$ is the converged loss, evaluated with infinite available data.

中 其中， $L(N, \infty)$ 是在可用数据无限的条件下评估得到的收敛损失。

EN This inequality and its comparison to the empirical data is displayed in Figure 16 in the appendix.

中 该不等式及其与实验数据的比较展示在附录的图 16 中。

EN In that figure, the values of S_{stop} and $L(N, D)$ are empirical (though S_{stop} is adjusted to mimic training at $B \gg B_{\text{crit}}$), while $L(N, \infty)$ is computed from the fit to $L(N, D)$ evaluated at $D = \infty$.

中 在该图中， S_{stop} 和 $L(N, D)$ 的值来自实验（不过，为了模拟在 $B \gg B_{\text{crit}}$ 下训练，对 S_{stop} 进行了校正），而 $L(N, \infty)$ 是根据对 $L(N, D)$ 的拟合在 $D = \infty$ 处计算得出的。

6 Optimal Allocation of the Compute Budget 计算预算的最优分配

EN We displayed the *empirical* trend of performance as a function of the computation used during training in the top-right of Figure 1.

中 我们在图 1 的右上方展示了性能如何随训练期间使用的计算量变化这一实验趋势。

EN However, this result involved training at a fixed batch size B , whereas we know that in fact we could train more efficiently⁸ by training at the batch size B_{crit} discussed in Section 5.1.

中 然而，这一结果涉及在固定 batch size B 下训练，而我们知道，事实上，采用第 5.1 节讨论的 batch size B_{crit} 进行训练可以提高效率⁹。

⁸One might ask why we did not simply train at B_{crit} in the first place. The reason is that it depends not only on the model but also on the target value of the loss we wish to achieve, and so is a moving target.

⁹人们或许会问，为什么我们一开始不直接在 B_{crit} 下训练。原因在于，它不仅取决于模型，还取决于我们希望达到的目标损失值，因而是一个不断变化的目标。

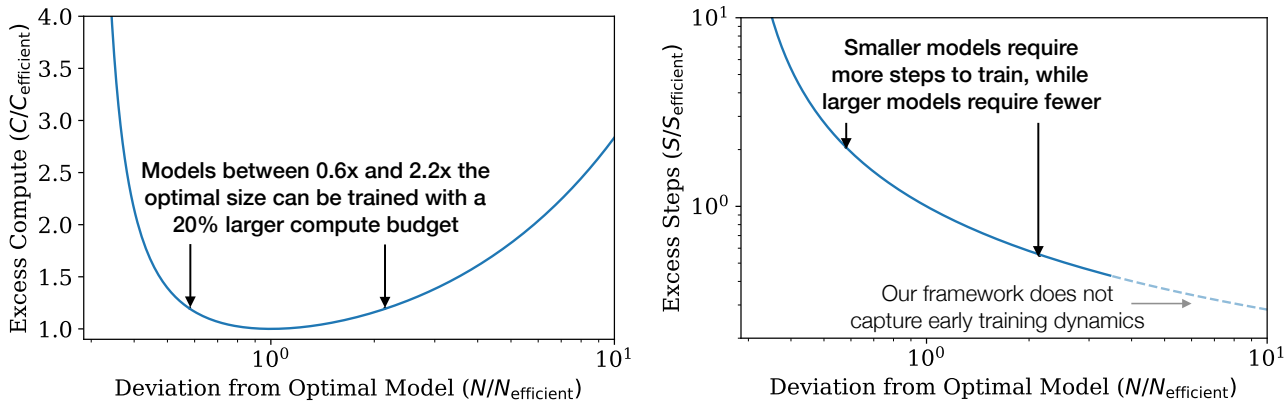


Figure / 图 12 **EN** Left: Given a fixed compute budget, a particular model size is optimal, though somewhat larger or smaller models can be trained with minimal additional compute.

中 左图：给定固定的计算预算，某个特定的模型规模是最优的，不过训练略大或略小的模型只需极少的额外计算量。

EN Right: Models larger than the compute-efficient size require fewer steps to train, allowing for potentially faster training if sufficient additional parallelism is possible.

中 右图：模型规模大于计算高效训练所对应的规模时，所需训练步数更少，因此，如果能够实现足够多的额外并行计算，训练或许可以更快。

EN Note that this equation should not be trusted for very large models, as it is only valid in the power-law region of the learning curve, after initial transient effects.

中 请注意，对于非常大的模型，不应相信该公式，因为它只在学习曲线的幂律区间内，即初始瞬态效应之后才有效。

EN Large and small values of the loss could have been achieved with fewer samples or fewer steps, respectively, and correcting for this inefficiency by standardizing to the critical batch size results in cleaner and more predictable trends.

中 较大和较小的损失值本可分别用更少的样本或更少的步数达到；通过将训练标准化至 critical batch size 来校正这种低效率，会得到更简洁、更可预测的趋势。

EN In this section we will adjust for this oversight.

中 在本节中，我们将校正这一疏忽。

EN More importantly, we will use the results of Section 5 to determine the optimal *allocation* of compute between model size N and the quantity of data processed during training, namely $2B_{\text{crit}}S_{\text{min}}$.

中 更重要的是，我们将使用第 5 节的结果，确定如何在模型规模 N 与训练期间处理的数据量（即 $2B_{\text{crit}}S_{\text{min}}$ ）之间最优地分配计算量。

EN We will determine this allocation both empirically and theoretically, by using the equation for $L(N, S_{\text{min}})$, and we will demonstrate that these methods agree.

中 我们将使用 $L(N, S_{\text{min}})$ 的公式，以实验和理论两种方式确定这一分配，并证明两种方法得到的结果一致。

6.1 Optimal Performance and Allocations

最优性能与分配

EN Let us first study the loss as a function of the optimally allocated compute from Equation (5.5).

中 首先来研究损失如何随公式 (5.5) 中最优分配的计算量变化。

EN The result is plotted in Figure 13, along with a power-law fit.

中 结果绘制在图 13 中，同时给出了幂律拟合。

EN We see that as compared to the compute plot of Figure 1, the new fit with C_{min} is somewhat improved.

中 我们看到，与图 1 中的计算量图相比，使用 C_{min} 的新拟合有所改善。

EN Given $L(C_{\text{min}})$, it is natural to ask for the optimal model size $N(C_{\text{min}})$ that provides the minimal loss with a given quantity of training compute.

中 给定 $L(C_{\text{min}})$ ，自然会想知道，在给定训练计算量下，能提供最小损失的最优模型规模 $N(C_{\text{min}})$ 是多少。

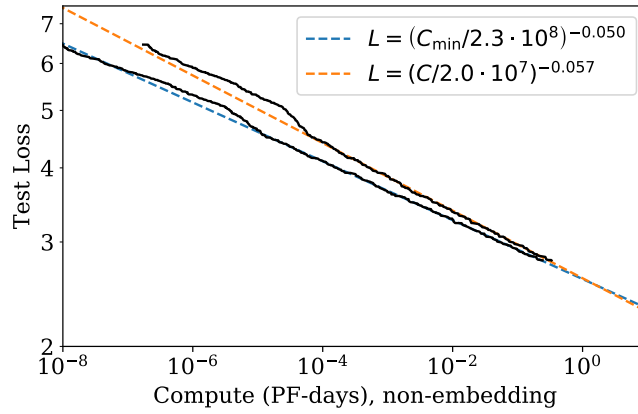


Figure / 图 13 *EN* When adjusting performance to simulate training far below the critical batch size, we find a somewhat altered power law for $L(C_{\min})$ when compared with the fully empirical results.

中 在对性能进行校正，以模拟远低于 critical batch size 的训练时，我们发现，与完全基于实验的结果相比， $L(C_{\min})$ 的幂律有所改变。

EN The conspicuous lump at 10^{-5} PF-days marks the transition from 1-layer to 2-layer networks; we exclude 1-layer networks in the power-law fits.

中 位于 10^{-5} PF-日处的显眼凸起标志着从 1 层网络到 2 层网络的转变；我们在幂律拟合中排除了 1 层网络。

EN It is the $L(C_{\min})$ trend that we expect to provide a reliable extrapolation for larger compute.

中 我们预计， $L(C_{\min})$ 趋势能够为更大计算量提供可靠的外推。

EN The optimal model size is shown in Figure 14.

中 最优模型规模展示在图 14 中。

EN We observe that $N(C_{\min})$ can be fit very well with a power-law

中 我们观察到， $N(C_{\min})$ 可以用幂律很好地拟合：

$$N(C_{\min}) \propto (C_{\min})^{0.73}. \quad (6.1)$$

EN In Figure 12, we show the effect of training models of sub-optimal sizes (see Appendix B.4).

中 在图 12 中，我们展示了训练非最优规模模型的影响（见附录 B.4）。

EN By definition $C_{\min} \equiv 6NB_{\text{crit}}S$, and so we can use $N(C_{\min})$ to extract further results.

中 根据定义， $C_{\min} \equiv 6NB_{\text{crit}}S$ ，因此可以使用 $N(C_{\min})$ 得出更多结果。

EN In particular, since prior fits show $B \propto L^{-4.8}$ and $L \propto C_{\min}^{-0.05}$, we can conclude that $B_{\text{crit}} \propto C_{\min}^{0.24}$.

中 特别是，由于先前的拟合表明 $B \propto L^{-4.8}$ 且 $L \propto C_{\min}^{-0.05}$ ，我们可以得出 $B_{\text{crit}} \propto C_{\min}^{0.24}$ 。

EN This leads us to conclude that the optimal number of steps will only grow very slowly with compute, as

中 由此可见，最优步数只会随计算量极其缓慢地增长，即

$$S_{\min} \propto (C_{\min})^{0.03}, \quad (6.2)$$

EN matching the empirical results in Figure 14.

中 这与图 14 中的实验结果一致。

EN In fact the measured exponent is sufficiently small that our results may even be consistent with an exponent of zero.

中 事实上，测得的指数非常小，以至于我们的结果甚至可能与零指数相符。

EN Thus we conclude that as we scale up language modeling with an optimal allocation of computation, we should predominantly increase the model size N , while simultaneously scaling up the batch size via $B \propto B_{\text{crit}}$ with negligible increase in the number of serial steps.

中 因此，我们得出结论：当以最优计算量分配来扩展语言建模时，应当主要增大模型规模 N ，同时通过 $B \propto B_{\text{crit}}$ 增大 batch size，而串行步数几乎无需增加。

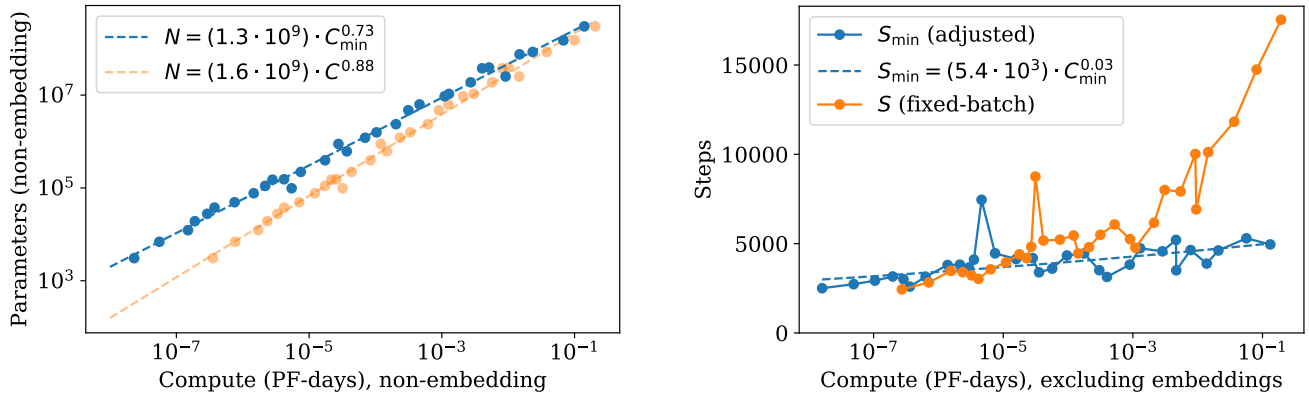


Figure / 图 14 **EN** Left: Each value of the compute budget $C_{\min}$ has an associated optimal model size N .

中 左图：计算预算 $C_{\min}$ 的每个取值都有一个相应的最优模型规模 N 。

EN Optimal model size grows very rapidly with $C_{\min}$, increasing by 5x for each 10x increase in compute.

中 最优模型规模随 $C_{\min}$ 极快增长：模型规模每增长 5 倍，对应的计算量增长 10 倍。

EN The number of data examples processed makes up the remainder of the increase, growing relatively modestly by only 2x.

中 已处理数据样本数构成其余的增量，增长幅度相对温和，仅为 2 倍。

EN Right: The batch-adjusted number of optimization steps also grows very slowly, if at all, meaning that most of the growth in data examples processed can be used for increased batch sizes.

中 右图：经批量校正的优化步数同样增长得非常缓慢，甚至可能根本没有增长，这意味着已处理数据样本数的大部分增长都可以用于增大 batch size。

EN Since compute-efficient training uses relatively few optimization steps, additional work on speeding up early training dynamics may be warranted.

中 由于计算高效训练使用的优化步数相对较少，或许有必要进一步研究如何加快训练早期的动力学过程。

6.2 Predictions from $L(N, S_{\min})$ 来自 $L(N, S_{\min})$ 的预测

EN The results for $L(C_{\min})$ and the allocations can be predicted from the $L(N, S_{\min})$ equation obtained in Section 5.

中 $L(C_{\min})$ 的结果和相应分配可以通过第 5 节所得的 $L(N, S_{\min})$ 公式来预测。

EN Given our equation for $L(N, S_{\min})$, we can substitute $S_{\min} = \frac{C_{\min}}{6NB}$ and then find the minimum of the loss as a function of N , while fixing the training compute.

中 给定 $L(N, S_{\min})$ 的公式，我们可以代入 $S_{\min} = \frac{C_{\min}}{6NB}$ ，然后在固定训练计算量的条件下，求损失关于 N 的最小值。

EN We carry out this procedure in detail in Appendix B, where we also provide some additional predictions.

中 我们在附录 B 中详细执行了这一过程，并在其中给出了一些额外预测。

EN For the loss as a function of training compute, we predict that

中 对于损失如何随训练计算量变化，我们预测

$$L(C_{\min}) = \left(\frac{C_c^{\min}}{C_{\min}} \right)^{\alpha_C^{\min}} \quad (6.3)$$

EN where

中 其中

$$\alpha_C^{\min} \equiv \frac{1}{1/\alpha_S + 1/\alpha_B + 1/\alpha_N} \approx 0.054 \quad (6.4)$$

EN in excellent agreement with the exponent of Figure 13.

中 这与图 13 中的指数高度吻合。

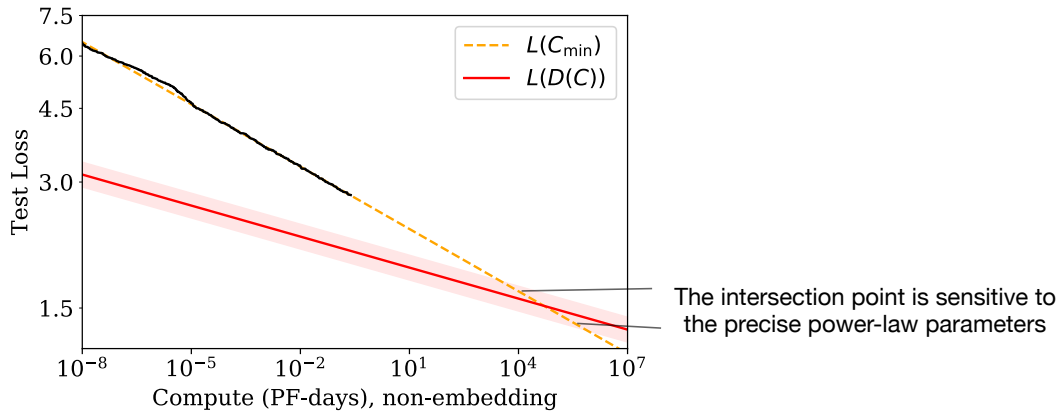


Figure / 图 15 EN Far beyond the model sizes we study empirically, we find a contradiction between our equations for $L(C_{\min})$ and $L(D)$ due to the slow growth of data needed for compute-efficient training.

中 在远远超出我们通过实验研究的模型规模时，我们发现，由于计算高效训练所需数据增长缓慢，关于 $L(C_{\min})$ 和 $L(D)$ 的公式之间出现了矛盾。

EN The intersection marks the point before which we expect our predictions to break down.

中 交点标出了我们预计预测会在其之前失效的位置。

EN The location of this point is highly sensitive to the precise exponents from our power-law fits.

中 该位置对幂律拟合所得的精确指数极为敏感。

EN We also predict that

中 我们还预测

$$N(C_{\min}) \propto (C_{\min})^{\alpha_{\min}^{\min}/\alpha_N} \approx (C_{\min})^{0.71} \quad (6.5)$$

EN which also matches the scaling of Figure 14 to within a few percent.

中 这同样在百分之几的范围内与图 14 的缩放关系相符。

EN Our scaling laws provide a predictive framework for the performance of language modeling.

中 我们的 scaling laws 为语言建模的性能提供了一个预测框架。

6.3 Contradictions and a Conjecture

矛盾与一个猜想

EN We observe no signs of deviation from straight power-law trends at large values of compute, data, or model size.

中 在计算量、数据或模型规模取较大值时，我们没有观察到偏离直线型幂律趋势的迹象。

EN Our trends must eventually level off, though, since natural language has non-zero entropy.

中 然而，我们的趋势最终必定趋于平缓，因为自然语言的熵非零。

EN Indeed, the trends for compute-efficient training described in this section already contain an apparent contradiction.

中 事实上，本节所述计算高效训练的趋势中已经包含了一个表面上的矛盾。

EN At scales several orders of magnitude above those documented here, the performance predicted by the $L(C_{\min})$ scaling law decreases below what should be possible given the slow growth in training data with compute.

中 当规模比本文记录的高出数个数量级时， $L(C_{\min})$ scaling law 预测的性能会下降到这样一种水平之下：考虑到训练数据随计算量增长得如此缓慢，这一水平本不可能达到。

EN This implies that our scaling laws must break down before this point, but we conjecture that the intersection point has a deeper meaning: it provides an estimate of the point at which Transformer language models reach maximal performance.

中 这意味着我们的 scaling laws 必定会在此之前失效，但我们猜想，该交点具有更深层的含义：它给出了 Transformer 语言模型达到最高性能之处的估计。

EN Since the amount of data used by compute-efficient training grows slowly with the compute budget, the performance predicted by $L(C_{\min})$ eventually hits a lower bound set by the $L(D)$ power law (see Figure 15).

中 由于计算高效训练使用的数据量随计算预算缓慢增长， $L(C_{\min})$ 预测的性能最终会触及由 $L(D)$ 幂律设定的下界（见图 15）。

EN Let us work this out in more detail.

中 下面对此作更详细的推导。

EN To keep overfitting under control, the results of Section 4 imply that we should scale the dataset size as

中 为了控制过拟合，第 4 节的结果表明，我们应按如下方式扩展数据集大小：

$$D \propto N^{0.74} \propto C_{\min}^{0.54} \quad (6.6)$$

EN where we have used the compute-efficient $N(C_{\min})$ from Figure 14.

中 这里，我们使用了图 14 中计算高效训练下的 $N(C_{\min})$ 。

EN Let us compare this to the data requirements of compute-efficient training.

中 下面将其与计算高效训练的数据需求进行比较。

EN If we train at the critical batch size (i.e. $C = 2C_{\min}$) and never re-use data during training, we find that data usage grows with compute as

中 如果在 critical batch size 下训练（即 $C = 2C_{\min}$ ），并且训练期间从不重复使用数据，那么数据使用量随计算量的增长关系为

$$D(C_{\min}) = \frac{2C_{\min}}{6N(C_{\min})} \approx (4 \times 10^{10} \text{ tokens}) (C_{\min}/\text{PF-Day})^{0.26} \quad (6.7)$$

EN This is the maximum rate at which the dataset size can productively grow with compute, since it means that we are only training for a single epoch.

中 这是数据集大小能够随计算量有效增长的最大速率，因为这意味着我们只训练一个轮次。

EN But it grows the dataset much more slowly than in Equation (6.6).

中 但与公式 (6.6) 相比，它使数据集增长得慢得多。

EN It appears to imply that compute-efficient training will eventually run into a problem with overfitting, even if the training process never re-uses any data!

中 它似乎意味着，即使训练过程从不重复使用任何数据，计算高效训练最终也会遇到过拟合问题！

EN According to Figure 1, we expect that when we are bottlenecked by the dataset size (ie by overfitting), the loss should scale as $L(D) \propto D^{-0.095}$.

中 根据图 1，我们预计，当瓶颈在于数据集大小（即受过拟合限制）时，损失应按 $L(D) \propto D^{-0.095}$ 缩放。

EN This implies that the loss would scale with compute as $L(D(C_{\min})) \propto C_{\min}^{-0.03}$ once we are data-limited.

中 这意味着，一旦受数据限制，损失将随计算量按 $L(D(C_{\min})) \propto C_{\min}^{-0.03}$ 缩放。

EN Once again, we have a contradiction, as this will eventually intersect with our prediction for $L(C_{\min})$ from Figure 13, where we found a scaling $L(C_{\min}) \propto C_{\min}^{-0.050}$.

中 我们再一次遇到了矛盾，因为这最终会与图 13 中关于 $L(C_{\min})$ 的预测相交；在该图中，我们发现的缩放关系是 $L(C_{\min}) \propto C_{\min}^{-0.050}$ 。

EN The intersection point of $L(D(C_{\min}))$ and $L(C_{\min})$ occurs at

中 $L(D(C_{\min}))$ 与 $L(C_{\min})$ 的交点位于

$$C^* \sim 10^4 \text{ PF-Days} \quad N^* \sim 10^{12} \text{ parameters}, \quad D^* \sim 10^{12} \text{ tokens}, \quad L^* \sim 1.7 \text{ nats/token} \quad (6.8)$$

EN though the numerical values are highly uncertain, varying by an order or magnitude in either direction depending on the precise values of the exponents from the power-law fits.

中 不过，这些数值具有很大的不确定性：取决于幂律拟合所得指数的精确取值，它们可能向任一方向变化一个数量级。

EN The most obvious interpretation is that our scaling laws break down at or before we reach this point, which is still many orders of magnitude away in both compute and model size.

中 最显而易见的解释是，我们的 scaling laws 会在达到这一点时或此前失效；而无论就计算量还是模型规模而言，这一点仍相距许多个数量级。

EN One might also conjecture that this intersection point has a deeper meaning.

中 也可以猜想，该交点具有更深层的含义。

EN If we cannot increase the model size beyond N^* without qualitatively different data requirements, perhaps this means that once we reach $C_{\min}^*$ and N^* , we have extracted all of the reliable information available in natural language data.

中 如果不在数据需求上发生质变，我们就无法将模型规模增大到 N^* 以上，那么也许这意味着，一旦达到 $C_{\min}^*$ 和 N^* ，我们就已经提取了自然语言数据中所有可靠的信息。

EN In this interpretation, L^* would provide a rough estimate for the entropy-per-token¹⁰ of natural language.

中 在这一解释下， L^* 将为自然语言的每 token 熵¹¹提供一个粗略估计。

EN In this scenario, we would expect the loss trend to level off at or before L^* .

中 在这种情形下，我们预计损失趋势会在 L^* 或此前趋于平缓。

EN We can guess at the functional form of $L(C_{\min})$ as it levels off by considering a version of our training dataset with added noise.

中 我们可以考虑在训练数据集中加入噪声的版本，以猜测 $L(C_{\min})$ 趋于平缓时的函数形式。

EN For example, we could append a random string of tokens to each context shown to the model to artificially boost the loss by a constant additive factor.

中 例如，可以在展示给模型的每段上下文后附加一个随机 token 串，从而人为地将损失提高一个恒定的加性因子。

EN Then, the distance from the noise floor $L - L_{\text{noise}}$ would be a more meaningful performance metric, with even a small decrease in this distance potentially representing a significant boost in qualitative performance.

中 这样，与噪声下限的距离 $L - L_{\text{noise}}$ 将成为更有意义的性能指标；即使该距离略有减小，也可能代表定性性能的显著提升。

EN Since the artificial noise would affect all of our trends equally, the critical point of 6.8 would not change (aside from the absolute value of L^*), and may be meaningful even if it occurs after the leveling off.

中 由于人为噪声会同等影响我们的所有趋势，6.8 的临界点不会改变（除了 L^* 的绝对值），即使它出现在趋势趋于平缓之后，也可能仍具有意义。

7 Related Work

相关工作

EN Power laws can arise from a wide variety of sources [THK18].

中 幂律可以由多种多样的来源产生 [THK18]。

EN Power-law scalings with model and dataset size in density estimation [Was06] and in random forest models [Bia12] may be connected with our results.

中 密度估计 [Was06] 和随机森林模型 [Bia12] 中随模型和数据集大小变化的幂律缩放关系可能与我们的结果有关。

EN These models suggest that power-law exponents may have a very rough interpretation as the inverse of the number of relevant features in the data.

中 这些模型表明，可以非常粗略地将幂律指数解释为数据中相关特征数量的倒数。

EN Some early [BB01, Goo01] work found power-law scalings between performance and dataset size.

中 一些早期工作 [BB01, Goo01] 发现了性能与数据集大小之间的幂律缩放关系。

EN More recent work [HNA⁺17, HAD19] also investigated scaling between model size and data size; their work is perhaps the closest to ours in the literature¹².

中 较新的工作 [HNA⁺17, HAD19] 还研究了模型规模与数据大小之间的缩放关系；在现有文献中，他们的工作或许与我们最为接近¹³。

EN Note, however, that [HNA⁺17] found super-linear scaling of dataset size with model size, whereas we find a sub-linear scaling.

中 但请注意，[HNA⁺17] 发现数据集大小相对于模型规模呈超线性缩放，而我们发现的是次线性缩放。

EN There are some parallels between our findings on optimal allocation of compute and [Kom19], including power-law learning curves.

中 我们关于计算量最优分配的发现与 [Kom19] 之间有一些相似之处，其中包括幂律学习曲线。

¹⁰Defining words using the `wc` utility, the WebText2 dataset has 1.4 tokens per word and 4.3 characters per token.

¹¹使用 `wc` 工具定义单词时，WebText2 数据集每个单词含 1.4 tokens，每个 token 含 4.3 个字符。

¹²After this work was completed, [RRBS19a] also appeared, which makes similar predictions for the dependence of loss on both model and dataset size.

¹³在本研究完成后，[RRBS19a] 也发表了，该研究对损失如何同时依赖于模型规模和数据集大小作出了类似预测。

EN EfficientNets [TL19] also appear to obey an approximate power-law relation between accuracy and model size.

中 EfficientNets [TL19] 似乎也遵循准确率与模型规模之间的近似幂律关系。

EN Very recent work [RRBS19b] studies scaling with both dataset size and model size for a variety of datasets, and fits an ansatz similar to ours.

中 最近的一项工作 [RRBS19b] 研究了多种数据集中数据集大小和模型规模同时变化时的缩放关系，并拟合了一种与我们类似的假设形式。

EN EfficientNet [TL19] advocates scaling depth and width exponentially (with different coefficients) for optimal performance of image models, resulting in a power-law scaling of width as a function of depth.

中 EfficientNet [TL19] 主张，为获得图像模型的最优性能，应以指数方式（采用不同系数）扩展深度和宽度，从而使宽度关于深度呈幂律缩放关系。

EN We find that for language models this power should be roughly one when scaling up (as width/depth should remain fixed).

中 我们发现，对于语言模型，在扩大规模时，该幂指数应当大致为一（因为宽度/深度应保持不变）。

EN But more importantly, we find that the precise architectural hyperparameters are unimportant compared to the overall scale of the language model.

中 但更重要的是，我们发现，相较语言模型的整体规模，具体的架构超参数并不重要。

EN In [VWB16] it was argued that deep models can function as ensembles of shallower models, which could potentially explain this finding.

中 [VWB16] 认为，深层模型可以充当较浅模型的集成，这可能解释这一发现。

EN Earlier work [ZK16] has compared width and depth, and found that wide ResNets can outperform deep ResNets on image classification.

中 更早的工作 [ZK16] 比较了宽度和深度，并发现宽 ResNets 在图像分类上的表现可以优于深 ResNets。

EN Some studies fix computation per data example, which tends to scale in proportion to the number of model parameters, whereas we investigate scaling with both model size and the quantity of training computation.

中 一些研究固定了每个数据样本的计算量，这一计算量往往与模型参数的数量成比例缩放；而我们研究的是模型规模和训练计算量同时变化时的缩放关系。

EN Various works [AS17, BHMM18] have investigated generalization in highly overparameterized models, finding a “jamming transition” [GJS⁺19] when the model size reaches the dataset size (this may require training many orders of magnitude beyond typical practice, and in particular does not use early stopping).

中 多项工作 [AS17, BHMM18] 研究了高度过参数化模型的泛化，发现在模型规模达到数据集大小时会出现一种“堵塞转变” [GJS⁺19]（这可能需要比典型实践多出许多个数量级的训练，尤其是不能使用 early stopping）。

EN We do not observe such a transition, and find that the necessary training data scales sublinearly in the model size.

中 我们没有观察到这样的转变，并发现所需训练数据相对于模型规模呈次线性缩放。

EN Expansions in the model size, particularly at large width [JGH18, LXS⁺19], may provide a useful framework for thinking about some of our scaling relations.

中 关于模型规模的展开，尤其是宽度很大时的展开 [JGH18, LXS⁺19]，可能为思考我们的某些缩放关系提供一个有用框架。

EN Our results on optimization, such as the shape of learning curves, can likely be explained using a noisy quadratic model, which can provide quite accurate predictions [ZLN⁺19] in realistic settings.

中 我们关于优化的结果，例如学习曲线的形状，很可能可以用含噪二次模型来解释；在现实情境中，这种模型能够作出相当准确的预测 [ZLN⁺19]。

EN Making this connection quantitative will require a characterization of the Hessian spectrum [Pap18, GKX19, GARD18].

中 要定量建立这一联系，需要刻画 Hessian 矩阵的谱 [Pap18, GKX19, GARD18]。

8 Discussion

讨论

- EN** We have observed consistent scalings of language model log-likelihood loss with non-embedding parameter count N , dataset size D , and optimized training computation $C_{\min}$, as encapsulated in Equations (1.5) and (1.6).
- 中 我们观察到，语言模型的对数似然损失随 non-embedding parameters 的数量 N 、数据集大小 D 和优化后的训练计算量 $C_{\min}$ 呈现出一致的缩放关系，如公式 (1.5) 和 (1.6) 所概括。
- EN** Conversely, we find very weak dependence on many architectural and optimization hyperparameters.
- 中 相反，我们发现它对许多架构和优化超参数的依赖非常弱。
- EN** Since scalings with N , D , $C_{\min}$ are power-laws, there are diminishing returns with increasing scale.
- 中 由于随 N , D , $C_{\min}$ 的缩放关系为幂律，随着规模增加，收益会递减。
- EN** We were able to precisely model the dependence of the loss on N and D , and alternatively on N and S , when these parameters are varied simultaneously.
- 中 当 N 与 D 同时变化，或者 N 与 S 同时变化时，我们能够精确地建立损失对它们的依赖模型。
- EN** We used these relations to derive the compute scaling, magnitude of overfitting, early stopping step, and data requirements when training large language models.
- 中 我们使用这些关系推导了训练大型语言模型时的计算量缩放、过拟合程度、early stopping 步数和数据需求。
- EN** So our scaling relations go beyond mere observation to provide a predictive framework.
- 中 因此，我们的缩放关系不止是单纯的观察，还提供了一个预测框架。
- EN** One might interpret these relations as analogues of the ideal gas law, which relates the macroscopic properties of a gas in a universal way, independent of most of the details of its microscopic constituents.
- 中 可以将这些关系理解为理想气体定律的类似物；该定律以普适的方式联系气体的宏观性质，而与其微观组成成分的大多数细节无关。
- EN** It is natural to conjecture that the scaling relations will apply to other generative modeling tasks with a maximum likelihood loss, and perhaps in other settings as well.
- 中 很自然可以猜想，这些缩放关系将适用于其他采用最大似然损失的生成建模任务，也可能适用于其他情境。
- EN** To this purpose, it will be interesting to test these relations on other domains, such as images, audio, and video models, and perhaps also for random network distillation.
- 中 为此，在图像、音频和视频模型等其他领域，或许也在随机网络蒸馏中检验这些关系，将会很有意义。
- EN** At this point we do not know which of our results depend on the structure of natural language data, and which are universal.
- 中 目前，我们尚不清楚哪些结果取决于自然语言数据的结构，哪些结果具有普适性。
- EN** It would also be exciting to find a theoretical framework from which the scaling relations can be derived: a ‘statistical mechanics’ underlying the ‘thermodynamics’ we have observed.
- 中 如果能找到一个可以从中推导缩放关系的理论框架，也会令人振奋：即为我们所观察到的“热力学”奠定基础的“统计力学”。
- EN** Such a theory might make it possible to derive other more precise predictions, and provide a systematic understanding of the limitations of the scaling laws.
- 中 这样的理论可能使我们能够推导出其他更精确的预测，并系统地理解 scaling laws 的局限。
- EN** In the domain of natural language, it will be important to investigate whether continued improvement on the loss translates into improvement on relevant language tasks.
- 中 在自然语言领域，研究损失的持续改善是否会转化为相关语言任务上的改善十分重要。
- EN** Smooth quantitative change can mask major qualitative improvements: “more is different”.
- 中 平滑的量变可能掩盖重大的质变：“多则异”。
- EN** For example, the smooth aggregate growth of the economy provides no indication of the specific technological developments that underwrite it.
- 中 例如，经济总量的平滑增长并不能揭示支撑这种增长的具体技术进步。
- EN** Similarly, the smooth improvements in language model loss may hide seemingly qualitative changes in capability.
- 中 类似地，语言模型损失的平滑改善可能隐藏着看似定性的能力变化。

EN Our results strongly suggest that larger models will continue to perform better, and will also be much more sample efficient than has been previously appreciated.

中 我们的结果强烈表明，更大的模型会继续表现得更好，而且样本效率也会比此前认识的更高。

EN Big models may be more important than big data.

中 大模型可能比大数据更重要。

EN In this context, further investigation into model parallelism is warranted.

中 在这种背景下，有必要进一步研究模型并行。

EN Deep models can be trained using pipelining [HCC⁺18], which splits parameters depth-wise between devices, but eventually requires increased batch sizes as more devices are used.

中 深层模型可以使用流水线方法训练 [HCC⁺18]，该方法沿深度方向将参数分配到不同设备之间，但随着使用的设备增多，最终会要求增大 batch size。

EN Wide networks on the other hand are more amenable to parallelization [SCP⁺18], since large layers can be split between multiple workers with less serial dependency.

中 另一方面，宽网络更适合并行化 [SCP⁺18]，因为大层可以被分割到多个工作节点之间，并且串行依赖更少。

EN Sparsity [CGRS19, GRK17] or branching (e.g. [KSH12]) may allow for even faster training of large networks through increased model parallelism.

中 稀疏性 [CGRS19, GRK17] 或分支（例如 [KSH12]）可能通过提高模型并行度，实现对大型网络更快的训练。

EN And using methods like [WRH17, WYL19], which grow networks as they train, it might be possible to remain on the compute-efficient frontier for an entire training run.

中 而采用 [WRH17, WYL19] 一类在训练过程中扩展网络的方法，或许可以在整个训练运行期间始终处于计算高效前沿。

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Appendices / 附录

A Summary of Power Laws

幂律总结

EN For easier reference, we provide a summary below of the key trends described throughout the paper.

中 为便于查阅，下面汇总了全文所述的关键趋势。

Parameters	Data	Compute	Batch Size	Equation
N	∞	∞	Fixed	$L(N) = (N_c/N)^{\alpha_N}$
∞	D	Early Stop	Fixed	$L(D) = (D_c/D)^{\alpha_D}$
Optimal	∞	C	Fixed	$L(C) = (C_c/C)^{\alpha_C}$ (naive)
N_{opt}	D_{opt}	$C_{\min}$	$B \ll B_{\text{crit}}$	$L(C_{\min}) = (C_c^{\min}/C_{\min})^{\alpha_C^{\min}}$
N	D	Early Stop	Fixed	$L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_D}$
N	∞	S steps	B	$L(N, S) = \left(\frac{N_c}{N} \right)^{\alpha_N} + \left(\frac{S_c}{S_{\min}(S, B)} \right)^{\alpha_S}$

Table / 表 4 **EN** Key trend equations

中 关键趋势方程

EN The empirical fitted values for these trends are:

中 这些趋势的经验拟合值为：

Power Law	Scale (tokenization-dependent)
$\alpha_N = 0.076$	$N_c = 8.8 \times 10^{13}$ params (non-embed)
$\alpha_D = 0.095$	$D_c = 5.4 \times 10^{13}$ tokens
$\alpha_C = 0.057$	$C_c = 1.6 \times 10^7$ PF-days
$\alpha_C^{\min} = 0.050$	$C_c^{\min} = 3.1 \times 10^8$ PF-days
$\alpha_B = 0.21$	$B_* = 2.1 \times 10^8$ tokens
$\alpha_S = 0.76$	$S_c = 2.1 \times 10^3$ steps

Table / 表 5 **EN** Key parameters to trend fits

中 趋势拟合的关键参数

EN The optimal parameters for compute efficient training are given by:

中 计算高效训练的最优参数如下：

Compute-Efficient Value	Power Law	Scale
$N_{\text{opt}} = N_e \cdot C_{\min}^{p_N}$	$p_N = 0.73$	$N_e = 1.3 \cdot 10^9$ params
$B \ll B_{\text{crit}} = \frac{B_*}{L^{1/\alpha_B}} = B_e C_{\min}^{p_B}$	$p_B = 0.24$	$B_e = 2.0 \cdot 10^6$ tokens
$S_{\min} = S_e \cdot C_{\min}^{p_S}$ (lower bound)	$p_S = 0.03$	$S_e = 5.4 \cdot 10^3$ steps
$D_{\text{opt}} = D_e \cdot C_{\min}^{p_D}$ (1 epoch)	$p_D = 0.27$	$D_e = 2 \cdot 10^{10}$ tokens

Table / 表 6 **EN** Trends for compute-efficient training

中 计算高效训练的趋势

B Empirical Model of Compute-Efficient Frontier

计算高效前沿的经验模型

EN Throughout this appendix all values of C , S , and α_C are adjusted for training at the critical batch size B_{crit} .

中 在本附录中, C , S , 和 α_C 的所有取值都已针对在 critical batch size B_{crit} 下的训练进行了调整。

EN We have left off the ‘adj’ label to avoid cluttering the notation.

中 为避免符号过于繁杂, 我们省略了 ‘adj’ 标记。

B.1 Defining Equations

定义方程

EN The power-law fit to the learning curves implies a simple prescription for compute-efficient training.

中 对学习曲线进行幂律拟合, 可以得到一种简单的计算高效训练方案。

EN In this appendix, we will derive the optimal performance, model size, and number of training steps as a function of the compute budget.

中 在本附录中, 我们将推导最优性能、模型规模和训练步数与计算预算之间的函数关系。

EN We start with the Equation (1.6), repeated here for convenience:

中 我们从方程 (1.6) 开始, 为方便起见在此重复如下:

$$L(N, S) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{S_c}{S}\right)^{\alpha_S}. \quad (\text{B.1})$$

EN Here, S represents the number of parameter updates when training **at the critical batch size** [MKAT18], which was defined in Equation (5.2)¹⁴:

中 这里, S 表示在 **critical batch size** 下训练时的参数更新次数 [MKAT18], critical batch size 在方程 (5.2)¹⁵ 中定义:

$$B(L) = \frac{B_*}{L^{1/\alpha_B}}. \quad (\text{B.2})$$

EN We would like to determine optimal training parameters for a fixed compute budget, so we replace $S = C / (6NB(L))$, where C is the number of FLOPs used in the training run:

中 我们希望确定固定计算预算下的最优训练参数, 因此作替换 $S = C / (6NB(L))$, 其中 C 是该次训练所使用的 FLOP 数:

$$L(N, C) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(6B_*S_c \frac{N}{L^{1/\alpha_B}C}\right)^{\alpha_S}. \quad (\text{B.3})$$

EN Now, we set $\partial_N L|_C = 0$ to find the condition for optimality:

中 现在, 我们令 $\partial_N L|_C = 0$, 以求得最优条件:

$$\begin{aligned} 0 &= \frac{\partial L}{\partial N}|_C \\ &= -\frac{\alpha_N}{N} \left(\frac{N_c}{N}\right)^{\alpha_N} + \frac{\alpha_S}{N} \left(6B_*S_c \frac{N}{L^{1/\alpha_B}C}\right)^{\alpha_S} \left(1 - 5 \frac{N}{L} \frac{\partial L}{\partial N}|_C\right) \\ \Rightarrow \frac{\alpha_N}{\alpha_S} \left(\frac{N_c}{N}\right)^{\alpha_N} &= \left(6B_*S_c \frac{N}{L^{1/\alpha_B}C}\right)^{\alpha_S} \end{aligned} \quad (\text{B.4})$$

EN Equation (B.3) and (B.4) together determine the compute-efficient frontier.

中 方程 (B.3) 和 (B.4) 共同确定了计算高效前沿。

B.2 Efficient Training

高效训练

¹⁴There is a slight ambiguity here: we can imagine training either at a constant batch size $B(L_{\text{target}})$, or we could instead train at a variable batch size $\tilde{B}(L)$, where $\tilde{B}$ is the instantaneous critical batch size (as opposed to B , which is the averaged version). These two prescriptions result in the same number of steps, so we can ignore this subtlety (see [MKAT18]).

¹⁵这里存在一点歧义: 我们可以设想在恒定 batch size $B(L_{\text{target}})$ 下训练, 也可以改为在可变 batch size $\tilde{B}(L)$ 下训练, 其中 $\tilde{B}$ 是瞬时 critical batch size (相对于 B , 后者是平均版本)。这两种方案产生的步数相同, 因此我们可以忽略这一细微差别 (见 [MKAT18])。

EN Now we assemble the implications of (B.3) and (B.4).

中 现在，我们汇总 (B.3) 和 (B.4) 的推论。

EN First, note that inserting (B.4) into (B.3) yields

中 首先，注意到将 (B.4) 代入 (B.3) 可得

$$L(N_{\text{eff}}(C), C) = \left(1 + \frac{\alpha_N}{\alpha_S}\right) L(N_{\text{eff}}, \infty), \quad (\text{B.5})$$

EN which implies that for compute-efficient training, we should train to a **fixed percentage** $\frac{\alpha_N}{\alpha_S} \approx 10\%$ above the converged loss.

中 这意味着，对于计算高效训练，我们应训练到比收敛损失高出固定百分比 $\frac{\alpha_N}{\alpha_S} \approx 10\%$ 的水平。

EN Next, let's determine how the optimal loss depends on the compute budget.

中 接下来，我们确定最优损失如何依赖于计算预算。

EN Eliminating N yields a power-law dependence of performance on compute:

中 消去 N ，可得到性能关于计算量的幂律依赖关系：

$$L(C) = \left(\frac{C_c}{C}\right)^{\alpha_C} \quad (\text{B.6})$$

EN where we defined

中 其中我们定义

$$\alpha_C = 1 / (1/\alpha_S + 1/\alpha_B + 1/\alpha_N) \approx 0.052 \quad (\text{B.7})$$

$$C_c = 6N_c B_* S_c \left(1 + \frac{\alpha_N}{\alpha_S}\right)^{1/\alpha_S + 1/\alpha_N} \left(\frac{\alpha_S}{\alpha_N}\right)^{1/\alpha_S}. \quad (\text{B.8})$$

EN Similarly, we can eliminate L to find $N(C)$:

中 类似地，我们可以消去 L ，求得 $N(C)$ ：

$$\frac{N(C)}{N_c} = \left(\frac{C}{C_c}\right)^{\alpha_C/\alpha_N} \left(1 + \frac{\alpha_N}{\alpha_S}\right)^{1/\alpha_N} \quad (\text{B.9})$$

EN and

中 以及

$$S(C) = \frac{C_c}{6N_c B_*} \left(1 + \frac{\alpha_N}{\alpha_S}\right)^{-1/\alpha_N} \left(\frac{C}{C_c}\right)^{\alpha_C/\alpha_S} \quad (\text{B.10})$$

B.3 Comparison to Inefficient 与低效训练的比较

EN Typically, researchers train models until they appear to be close to convergence.

中 通常，研究人员会训练模型，直到它们看起来接近收敛。

EN In this section, we compare the efficient training procedure described above to this more typical setup.

中 在本节中，我们将上述高效训练过程与这种更为典型的设置进行比较。

EN We define a the convergence factor f as the percent deviation from the converged loss:

中 我们将收敛因子 f 定义为相对于收敛损失的百分比偏差：

$$L(N, C) = (1 + f) L(N, \infty). \quad (\text{B.11})$$

EN For compute-efficient training we have $f = \alpha_N / \alpha_S \approx 10\%$ from the previous section, but researchers typically use a much smaller value.

中 由上一节可知，对于计算高效训练， $f = \alpha_N / \alpha_S \approx 10\%$ ，但研究人员通常使用小得多的值。

EN Here, we choose $f' = 2\%$ as an estimate.

中 这里，我们选择 $f' = 2\%$ 作为估计值。

EN For a fixed value of the loss, we predict:

中 对于固定的损失值，我们预测：

$$\frac{N_f}{N_{f'}} = \left(\frac{1+f}{1+f'} \right)^{1/\alpha_N} \approx 2.7 \quad (\text{B.12})$$

$$\frac{S_f}{S_{f'}} = \left(\frac{1+\frac{1}{f}}{1+\frac{1}{f'}} \right)^{1/\alpha_S} \approx 0.13 \quad (\text{B.13})$$

$$\frac{C_f}{C_{f'}} = \frac{N_f}{N_{f'}} \frac{S_f}{S_{f'}} \approx 0.35 \quad (\text{B.14})$$

EN So that compute-efficient training uses 7.7x fewer parameter updates, 2.7x more parameters, and 65% less compute to reach the same loss.

中 因此，为达到相同的损失，计算高效训练所需的参数更新次数缩减 7.7 倍、参数数量增至 2.7 倍，并少用 65% 的计算量。

B.4 Suboptimal Model Sizes

次优模型规模

EN We can solve A.1 to find an expression for the amount of compute needed to reach a given value of the loss L with a model of size N :

中 我们可以求解 A.1，得到达到给定损失值 L 时，规模为 N 的模型所需计算量的表达式：

$$C(N, L) = \left(6B_* S_c \frac{N}{L^{1/\alpha_B}} \right) \left(L - \left(\frac{N_c}{N} \right)^{\alpha_N} \right)^{-1/\alpha_S}. \quad (\text{B.15})$$

EN Using A.6 and A.9, we can eliminate L in favor of $N_{\text{eff}}(L)$, the model size which reaches L most efficiently.

中 利用 A.6 和 A.9，我们可以消去 L ，改用 $N_{\text{eff}}(L)$ 表示，其中后者是以最高效率达到 L 的模型规模。

EN From there, we find an expression for the excess compute needed as a consequence of using a suboptimal model size:

中 由此，我们得到因使用次优模型规模而需要的额外计算量的表达式：

$$\frac{C(N, N_{\text{eff}})}{C(N_{\text{eff}}, N_{\text{eff}})} = \frac{N}{N_{\text{eff}}} \left[1 + \frac{\alpha_S}{\alpha_N} \left(1 - \left(\frac{N_{\text{eff}}}{N} \right)^{\alpha_N} \right) \right]^{-1/\alpha_S}. \quad (\text{B.16})$$

EN The result is shown in Figure X.

中 结果如图 X 所示。

EN Models between 0.6x and 2.2x the optimal size can be used with only a 20% increase in compute budget.

中 可以使用规模介于最优规模 0.6 倍和 2.2 倍之间的模型，而计算预算仅增加 20%。

EN Using a smaller model is useful when accounting for the cost inference.

中 考虑 inference 成本时，使用较小的模型很有价值。

EN A larger model can be trained the the same level of performance in fewer steps, allowing for more parallelism and faster training if sufficient hardware is available (see Figure Y):

中 更大的模型可以用更少的步数训练到相同的性能水平，从而在硬件充足时实现更高的并行度和更快的训练（见图 Y）：

$$\frac{S(N, N_{\text{eff}})}{S(N_{\text{eff}}, N_{\text{eff}})} = \left[1 + \frac{\alpha_S}{\alpha_N} \left(1 - \left(\frac{N_{\text{eff}}}{N} \right)^{\alpha_N} \right) \right]^{-1/\alpha_S}. \quad (\text{B.17})$$

EN A 2.2x larger model requires 45% fewer steps at a cost of 20% more training compute.

中 规模为 2.2 倍的模型所需步数减少 45%，代价是训练计算量增加 20%。

EN Note that this equation should not be trusted for very large models, as it is only valid in the power-law region of the learning curve after initial transient effects.

中 请注意，对于非常大的模型，不应信赖这个方程，因为它仅在学习曲线经过初始瞬态效应之后的幂律区域内有效。

C Caveats

注意事项

EN In this section we list some potential caveats to our analysis.

中 本节列出了我们分析中一些潜在的注意事项。

- **EN** At present we do not have a solid theoretical understanding for any of our proposed scaling laws.

中 目前，对于我们提出的所有 scaling laws，我们都缺乏扎实的理论理解。

EN The scaling relations with model size and compute are especially mysterious.

中 模型规模和计算量的缩放关系尤其难以解释。

EN It may be possible to understand scaling at very large D holding model size fixed [AS17], and also the shape of learning curves late in training, by modeling the loss with a noisy quadratic.

中 通过用带噪二次函数对损失建模，或许可以理解在模型规模固定、 D 非常大时的缩放关系 [AS17]，以及训练后期学习曲线的形状。

EN But the scaling with D at very large model size still remains mysterious.

中 但在模型规模非常大时随 D 的缩放关系仍然难以解释。

EN Without a theory or a systematic understanding of the corrections to our scaling laws, it's difficult to determine in what circumstances they can be trusted.

中 如果没有理论，或无法系统理解对 scaling laws 的修正，就很难确定在什么情况下可以信赖它们。

- **EN** We are not especially confident in the prediction of $B_{\text{crit}}(L)$ for values of the loss far outside the range we have explored.

中 对于远在我们所探索范围之外的损失值，我们并不特别确信对 $B_{\text{crit}}(L)$ 的预测。

EN Changes in B_{crit} could have a significant impact on trade-offs between data parallelism and the number of serial training steps required, which would have a major impact on training time.

中 B_{crit} 的变化可能会显著影响数据并行度与所需串行训练步数之间的权衡，进而对训练时间产生重大影响。

- **EN** We did not thoroughly investigate the small data regime, and our fits for $L(N, D)$ were poor for the smallest values of D (where an epoch corresponded to only 40 steps).

中 我们没有彻底研究小数据情形，并且我们对 $L(N, D)$ 的拟合在最小的 D 值时效果较差（此时一个轮次仅对应 40 步）。

EN Furthermore, we did not experiment with regularization and data augmentation.

中 此外，我们没有试验正则化和数据增强。

EN Improvements in these could alter our results, quantitatively or qualitatively.

中 对这些方法的改进可能会从定量或定性上改变我们的结果。

- **EN** We used the estimated training compute $C \approx 6NBS$, which did not include contributions proportional to n_{ctx} (see Section 2.1).

中 我们使用了估算的训练计算量 $C \approx 6NBS$ ，其中没有包含与 n_{ctx} 成正比的贡献（见第 2.1 节）。

EN So our scalings with compute may be confounded in practice in the regime of very large n_{ctx} , specifically where $n_{\text{ctx}} \gtrsim 12d_{\text{model}}$.

中 因此，在 n_{ctx} 非常大的区域，特别是 $n_{\text{ctx}} \gtrsim 12d_{\text{model}}$ 时，我们关于计算量的缩放关系在实践中可能会受到混杂因素的影响。

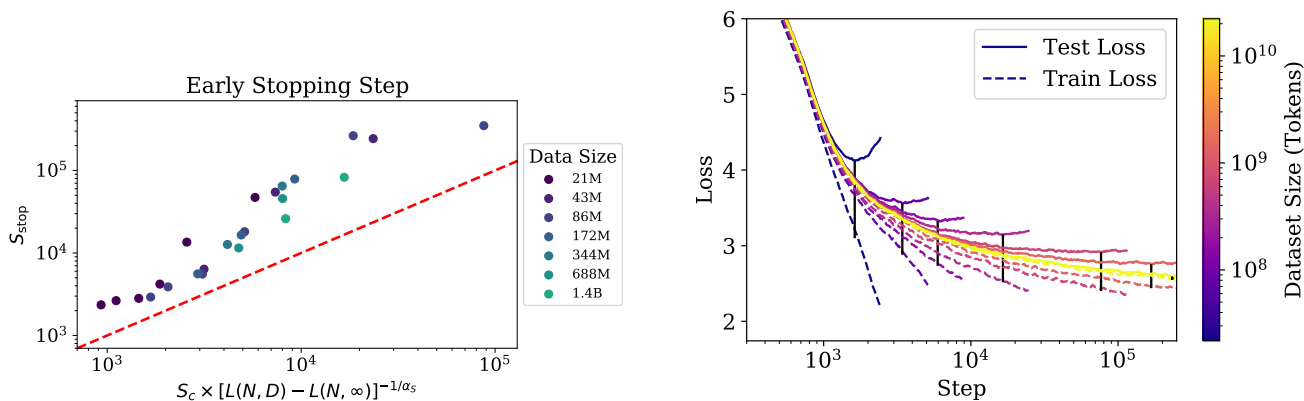


Figure / 图 16 **EN** Left: We characterize the step on which early stopping occurs, as a function of the extent of overfitting.

中 左: 我们刻画了 early stopping 发生的步数与过拟合程度之间的函数关系。

EN The red line indicates a lower bound for early stopping that is derived in Section 5.3.

中 红线表示第 5.3 节中推导出的 early stopping 下界。

EN Right: We display train and test loss for a series of 300M parameter models trained on different sized dataset sub-samples.

中 右: 我们展示了一系列 3 亿参数模型的训练损失和测试损失, 这些模型在不同规模的数据集子样本上训练。

EN The test loss typically follows that of a run done with unrestricted data until diverging.

中 测试损失通常会跟随使用无限制数据的训练运行, 直到两者出现偏离。

EN Note that the degree of overfitting (as compared to the infinite data limit) is significantly overestimated by $L_{\text{test}} - L_{\text{train}}$ (denoted by a black bar for each run).

中 请注意, $L_{\text{test}} - L_{\text{train}}$ 会显著高估过拟合程度 (与无限数据极限相比), 图中每次训练都用黑色条标出了这一差值。

- **EN** We tuned learning rates, and we experimented with learning rate schedules.

中 我们调节了 learning rate, 并试验了 learning-rate schedule。

EN But we may have neglected to tune some hyperparameter (e.g. initialization scale or momentum) that have an important effect on scaling.

中 但我们可能忽略了对某些会显著影响缩放的超参数 (例如初始化尺度或动量) 进行调节。

- **EN** The optimal choice of learning rate is sensitive to the target loss.

中 learning rate 的最优选择对目标损失很敏感。

EN When training close to convergence, it may be necessary to use a smaller learning rate to avoid divergences.

中 当训练接近收敛时, 可能需要使用较小的 learning rate 来避免发散。

EN But when conducting a short training run (eg due to compute limitations), it may be possible to use a larger learning rate.

中 但在进行短时间训练时 (例如由于计算量限制), 或许可以使用更大的 learning rate。

EN We did not experiment with higher learning rates for training runs that did not proceed to convergence.

中 对于未进行到收敛的训练, 我们没有试验更高的 learning rate。

D Supplemental Figures

补充图

D.1 Early Stopping and Test vs Train

Early Stopping 以及测试与训练的比较

EN In section 5.3 we described the result shown in Figure 16, which provides a prediction for a lower bound on the early stopping step.

中 在第 5.3 节中, 我们描述了图 16 所示的结果, 该结果预测了 early stopping 步数的下界。

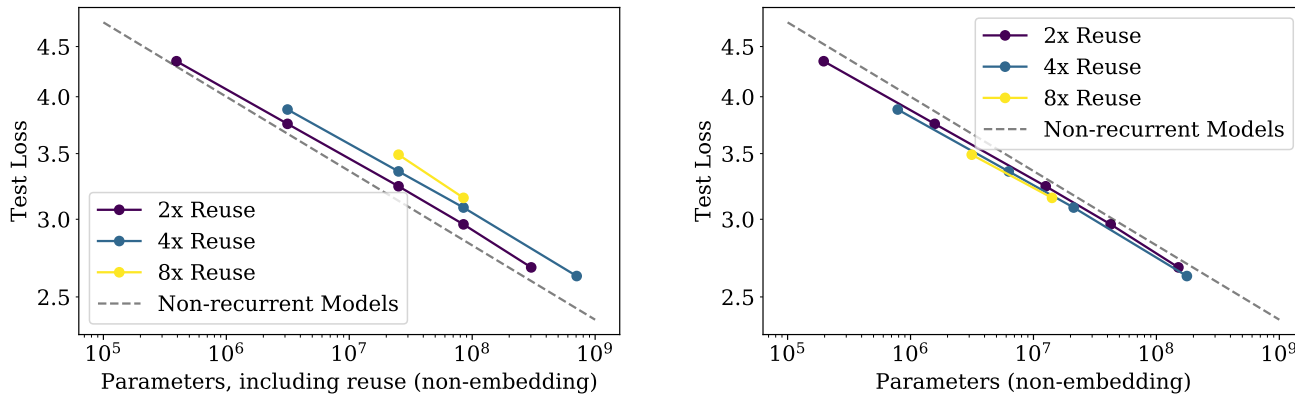


Figure / 图 17 *EN* We compare recurrent Transformers [DGV⁺18], which re-use parameters, to standard Transformers.

中 我们将复用参数的循环 Transformer [DGV⁺18] 与标准 Transformer 进行比较。

EN Recurrent Transformers perform slightly better when comparing models with equal parameter count, but slightly worse when accounting for reuse and comparing per FLOP.

中 在比较参数量相同的模型时，循环 Transformer 的表现略好；但考虑参数复用并按每 FLOP 比较时，其表现略差。

EN We also show the train and test loss for a given model size when training on different sized datasets.

中 我们还展示了在不同规模的数据集上训练时，给定模型规模对应的训练损失和测试损失。

D.2 Universal Transformers 通用 Transformer

EN We compare the performance of standard Transformers to recurrent Transformers [DGV⁺18] in Figure 17.

中 我们比较了标准 Transformer 与循环 Transformer [DGV⁺18] 的性能，结果见图 17。

EN These models re-use parameters, and so perform slightly better as a function of N , but slightly worse as a function of compute C .

中 这些模型会复用参数，因此按 N 衡量时表现略好，但按计算量 C 衡量时表现略差。

EN We include several different possibilities for parameter re-use.

中 我们纳入了几种不同的参数复用方式。

D.3 Batch Size Batch Size

EN We measure the critical batch size using the data displayed in figure 18.

中 我们使用图 18 所示的数据测量 critical batch size。

EN This made it possible to estimate $B_{\text{crit}}(L)$ in figure 10.

中 由此可以估算 $B_{\text{crit}}(L)$ ，如图 10 所示。

D.4 Sample Efficiency vs Model Size 样本效率与模型规模

EN It is easy to see from figure 2 that larger models train faster, and are therefore more sample efficient.

中 从图 2 中很容易看出，较大的模型训练得更快，因此样本效率也更高。

EN We provide another way of looking at this phenomenon in figure 19, which shows when different models reach various fixed values of the loss.

中 我们在图 19 中提供了观察这一现象的另一种方式，该图展示了不同模型何时达到各个固定的损失值。

D.5 Context Dependence 上下文依赖性

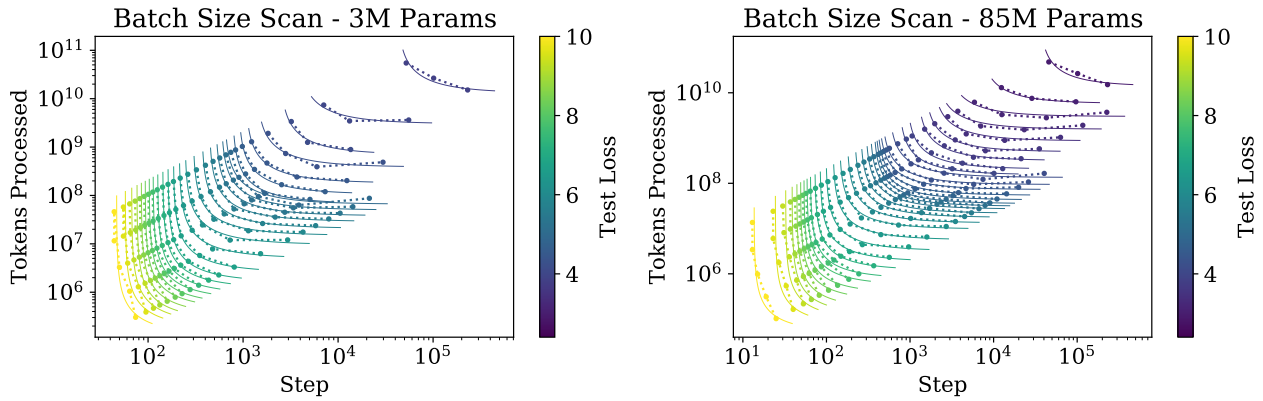


Figure / 图 18 EN These figures demonstrate fits to Equation (5.1) for a large number of values of the loss L , and for two different Transformer model sizes.

中 这些图展示了在损失 L 的大量不同取值下，针对两种不同规模的 Transformer 模型对方程 (5.1) 所做的拟合。

EN These fits were used to measure $B_{\text{crit}}(L)$ for Figure 10.

中 这些拟合用于测量 $B_{\text{crit}}(L)$ ，以绘制图 10。

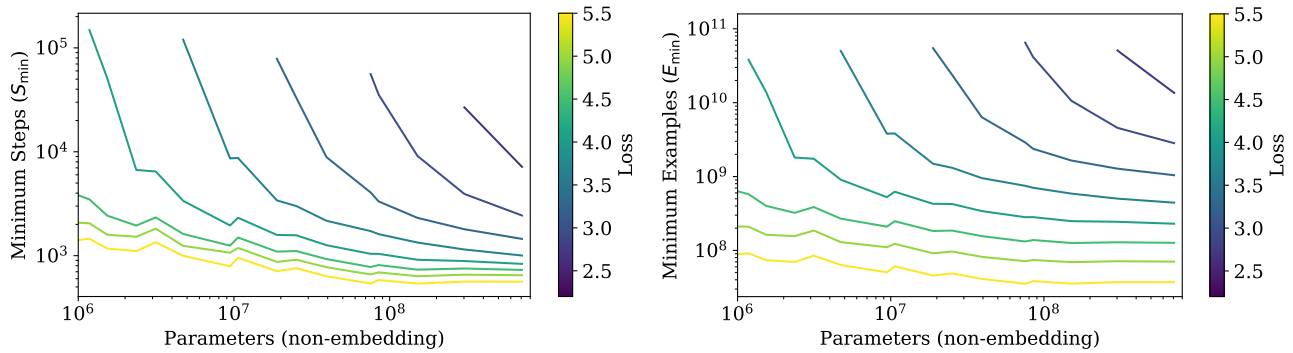


Figure / 图 19 EN The number of minimum serial steps needed to reach any fixed value of the test loss decreases precipitously with model size.

中 达到任意固定测试损失值所需的最少串行步数会随模型规模增大而急剧减少。

EN Sample efficiency (show here for training far below the critical batch size) improves greatly as well, improving by a factor of almost 100 when comparing the smallest possible model to a very large one.

中 样本效率（这里展示的是远低于 critical batch size 的训练）也有大幅提高；将最小的可行模型与非常大的模型进行比较时，提升幅度接近 100 倍。

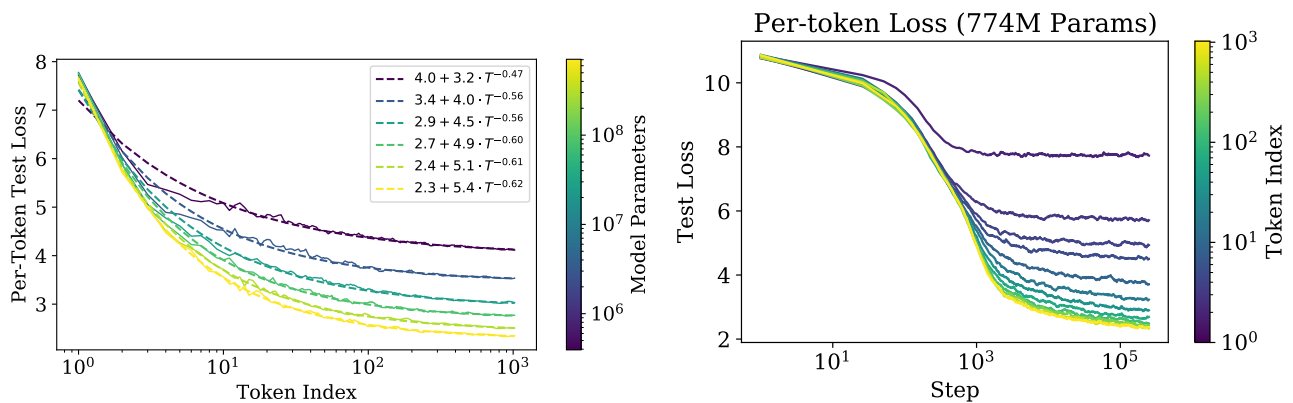


Figure / 图 20 EN This figure provides information about the performance per token as a function of model size and training time.

中 此图提供了每个 token 的性能如何随模型规模和训练时间变化的信息。

EN **Left:** Loss per token as a function of its position T in the 1024-token context.

中 左：每个 token 的损失与其在 1024-token 上下文中的位置 T 之间的函数关系。

EN Loss scales predictably as a power-law in T .

中 损失以可预测的方式随 T 呈幂律缩放。

EN **Right:** Test loss per token as a function of training step.

中 右：每个 token 的测试损失与训练步数之间的函数关系。

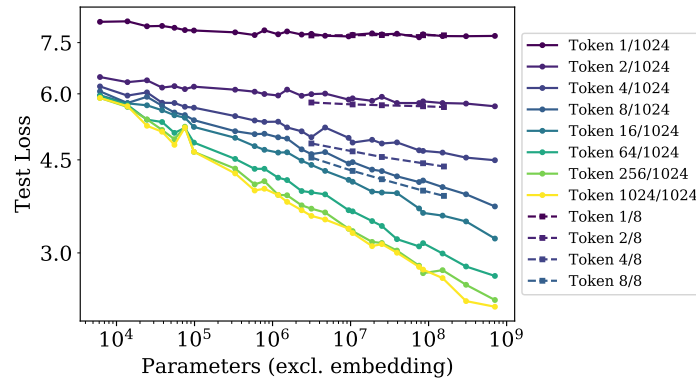


Figure / 图 21 *EN* In addition to the averaged loss, individual tokens within the 1024-token context also improve smoothly as model size increases.

除了平均损失外，1024-token 上下文中的各个 token 也会随着模型规模增大而平稳改善。

EN Training runs with shorter context $n_{\text{ctx}} = 8$ (dashed lines) perform better on early tokens, since they can allocate all of their capacity to them.

使用较短上下文 $n_{\text{ctx}} = 8$ 训练的模型（虚线）在较早的 token 上表现更好，因为它们可以将全部容量分配给这些 token。

EN The trends for loss as a function of model size are displayed for different tokens in the context in Figure 21.

图 21 展示了上下文中不同 token 对应的损失随模型规模变化的趋势。

EN We see that models trained on $n_{\text{ctx}} = 1024$ show steady improvement with model size on all but the first token.

我们看到，在 $n_{\text{ctx}} = 1024$ 上训练的模型，除第一个 token 外，其他所有 token 上的性能都会随模型规模增大而稳步改善。

EN Fixing model size, it appears that the loss scales as a power-law as a function of position T in the context, see Figure 20.

固定模型规模后，损失似乎会随上下文中的位置 T 呈幂律缩放，见图 20。

EN This may be a consequence of underlying power-law correlations in language [EP94, ACDE12, LT16], or a more general feature of the model architecture and optimization.

这可能是语言中底层幂律相关性的结果 [EP94, ACDE12, LT16]，也可能是模型架构和优化更普遍的一项特征。

EN It provides some suggestion for the potential benefits (or lack thereof) from training on larger contexts.

这为使用更大上下文训练可能带来的益处（或缺乏益处）提供了一些提示。

EN Not only do larger models converge to better performance at $T = 1024$, but they also improve more quickly at early tokens, suggesting that larger models are more efficient at detecting patterns with less contextual information.

较大的模型不仅在 $T = 1024$ 处收敛到更好的性能，而且在较早的 token 上改进得也更快，这表明较大的模型能以更少的上下文信息更高效地检测模式。

EN In the right-hand plot we show how per-token performance varies for a fixed model as a function of the training step.

在右图中，我们展示了固定模型的 per-token 性能如何随训练步数变化。

EN The model begins by learning short-range information, and only learns longer-range correlations later in training.

模型首先开始学习短程信息，直到训练后期才会学习长程相关性。

EN We have also included models trained with a tiny context $n_{\text{ctx}} = 8$ in order to compare with our longer context models.

我们还纳入了使用极小上下文 $n_{\text{ctx}} = 8$ 训练的模型，以便与较长上下文模型进行比较。

EN Even modestly sized models trained on $n_{\text{ctx}} = 8$ can dominate our largest $n_{\text{ctx}} = 1024$ models on very early tokens.

即使是规模适中的、在 $n_{\text{ctx}} = 8$ 上训练的模型，在非常早期的 token 上也能胜过我们最大的 $n_{\text{ctx}} = 1024$ 模型。

EN This also suggests that further improvements should be possible with much larger models trained on large contexts.

这也表明，使用长上下文训练的、规模大得多的模型，应该还能取得进一步改进。

D.6 Learning Rate Schedules and Error Analysis

learning-rate schedule 与误差分析

EN We experimented with a variety of learning rates and schedules.

我们试验了多种 learning rate 和调度。

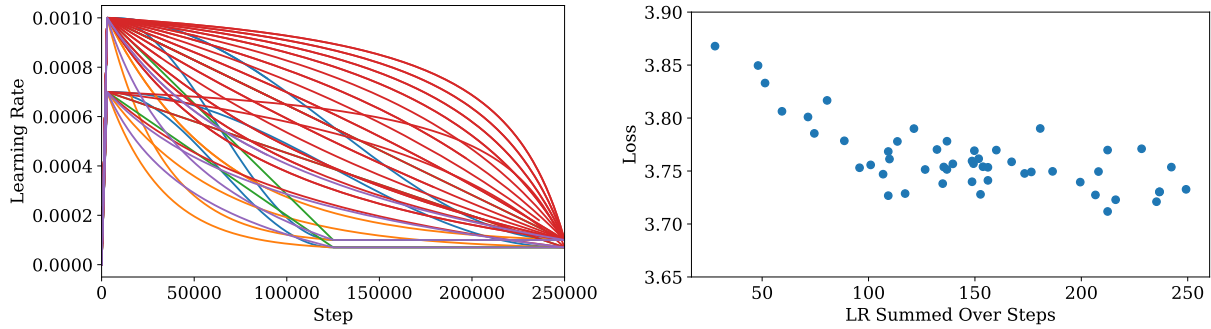


Figure / 图 22 *EN* We test a variety of learning rate schedules including cosine decay, linear decay, as well as other faster/slower decays schedules on a 3 million parameter model, shown on the left.

中 我们在一个 300 万参数模型上测试了多种 learning-rate schedule，包括余弦衰减、线性衰减，以及其他更快或更慢的衰减调度，如左图所示。

EN For these experiments we do not decay to zero, since we find that this tends to give a fixed improvement close to the end of training.

中 在这些实验中，我们没有将 learning rate 衰减到零，因为我们发现，这往往会在接近训练结束时带来一个固定的改进。

EN We find that, as long as the learning rate is not too small and does not decay too quickly, performance does not depend strongly on learning rate.

中 我们发现，只要 learning rate 不是过小，并且衰减速度不是过快，性能就不会强烈依赖于 learning rate。

EN Run-to-run variation is at the level of 0.05 in the loss, so averaging multiple runs is necessary to validate performance changes smaller than this level.

中 不同训练运行之间的损失变化约为 0.05，因此，要验证小于这一幅度的性能变化，就必须对多次训练运行取平均。

EN A host of schedules and resulting test performances for a small language model are plotted in Figure 22.

中 图 22 绘制了一个小型语言模型所采用的许多种调度及其相应测试性能。

EN We conclude that the choice of learning rate schedule is mostly irrelevant, as long as the total summed learning rate is sufficiently large, and the schedule includes a warmup period and a final decay to near-vanishing learning rate.

中 我们的结论是，只要 learning rate 总和足够大，并且调度包含预热阶段以及最终衰减至接近于零的 learning rate，那么 learning-rate schedule 的选择基本无关紧要。

EN Variations among schedules appear to be statistical noise, and provide a rough gauge for the scale of variation between different training runs.

中 不同调度之间的变化似乎是统计噪声，可以粗略衡量不同训练运行之间的变化幅度。

EN Experiments on larger models suggest that the variation in the final test loss between different random seeds is roughly constant in magnitude for different model sizes.

中 在较大模型上的实验表明，对于不同模型规模，不同随机种子之间最终测试损失的变化幅度大致恒定。

EN We found that larger models require a smaller learning rate to prevent divergence, while smaller models can tolerate a larger learning rate.

中 我们发现，较大的模型需要较小的 learning rate 来防止发散，而较小的模型则可以承受较大的 learning rate。

EN To implement this, the following rule of thumb was used for most runs:

中 为实现这一点，大多数训练运行采用了以下经验法则：

$$\text{LR}(N) \approx 0.003239 + -0.0001395 \log(N) \quad (\text{D.1})$$

EN We expect that this formula could be improved.

中 我们预计这个公式还可以改进。

EN There may be a dependence on network width, likely set by the initialization scale.

中 它可能依赖于网络宽度，而这很可能由初始化尺度决定。

EN The formula also breaks down for $N > 10^{10}$ parameters.

中 当参数量 $N > 10^{10}$ 时，该公式也会失效。

EN Nevertheless, we found that it works sufficiently well for the models we considered.

中 尽管如此，我们发现它对所考虑的模型已足够有效。

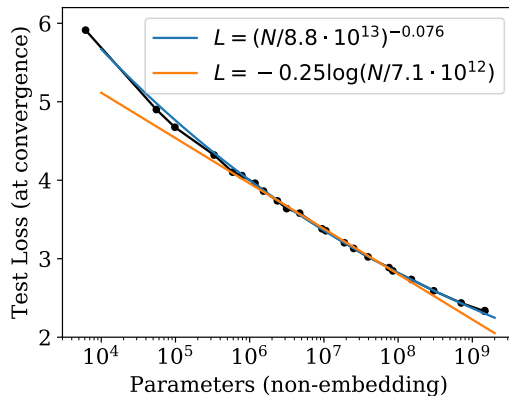


Figure / 图 23 *EN* The trend for performance as a function of parameter count, $L(N)$, is fit better by a power law than by other functions such as a logarithm at a qualitative level.

中 在定性层面上，以参数量为自变量的性能趋势 $L(N)$ 用幂律拟合比用对数等其他函数拟合得更好。

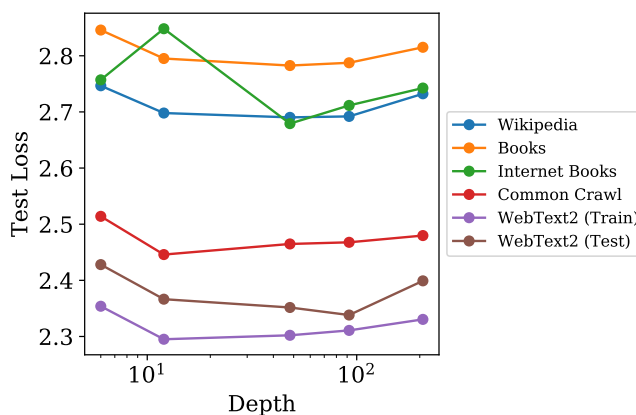


Figure / 图 24 *EN* We show evaluations on a series of datasets for models with approximately 1.5 Billion parameters.

中 我们展示了约有 15 亿个参数的模型在一系列数据集上的评估结果。

EN We observe no effect of depth on generalization; generalization performance depends primarily on training distribution performance.

中 我们没有观察到深度对泛化的影响；泛化性能主要取决于训练分布上的性能。

EN The 12-layer model overfit the Internet Books dataset and we show the early-stopped performance; we have not seen this surprising result in other experiments.

中 12 层模型在互联网图书数据集上发生了过拟合，因此我们展示了 early-stopped 性能；我们在其他实验中没有看到过这一令人意外的结果。

D.7 Fit Details and Power Law Quality

拟合细节与幂律质量

EN We experimented with a number of functional forms for the fits to $L(N)$, $L(C)$, and $L(D)$; the power-law fits were qualitatively much more accurate than other functions such as logarithms (see Figure 23).

中 我们尝试了多种函数形式来拟合 $L(N)$, $L(C)$ 和 $L(D)$ ；与对数等其他函数相比，幂律拟合在定性上准确得多（见图 23）。

EN For $L(C)$, we do not include small models with only 1 layer in the fit, as the transition from 1 to 2 layers causes a noticeable lump in the data.

中 对于 $L(C)$ ，我们没有在拟合中纳入仅有 1 层的小型模型，因为从 1 层过渡到 2 层会使数据出现明显的凸起。

EN For $L(N)$ we also do not include very small models with only 1 layer in the fit, and we exclude the largest models that have not trained fully to convergence.

中 对于 $L(N)$ ，我们同样没有在拟合中纳入仅有 1 层的极小模型，并且排除了尚未完全训练至收敛的最大模型。

EN Fit parameters change marginally if we do include them, and the trend extrapolates well in both directions regardless.

中 即使将它们纳入，拟合参数的变化也很小，并且无论是否纳入，该趋势在两个方向上都能很好地外推。

D.8 Generalization and Architecture

泛化与架构

EN In figure 24 we show that generalization to other data distributions does not depend on network depth when we hold the total parameter count fixed.

中 在图 24 中，我们表明，当总参数量固定时，对其他数据分布的泛化不依赖于网络深度。

EN It seems to depend only on the performance on the training distribution.

中 它似乎只取决于训练分布上的性能。

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